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Technical Report No. 54-6

Formalized Risk Assessment for Excipients

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Technical Report No. 54-6

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Foreword

Joint Initiative between IPEC Federation and PDA

In March 2018, the IPEC Federation and PDA announced their first joint initiative. Both associations signed a memorandum of understanding to collaborate on the development of a joint technical report on excipient good manufacturing practices risk assessment in response to input from their respective memberships. Prior to the collaboration announcement, however, both parties were active on the topic of quality risk management. PDA published the *PDA Technical Report No. 54* series. In March 2016, IPEC Europe published the *How-To Document – A Guide to Support Manufacturing Authorization Holders (MAH) in their Compliance with the European Commission Guidelines on Risk Assessment for Excipients (2015/C 95/02)*. Subsequently, in May 2017, IPEC-Americas and IPEC Europe jointly published the *Risk Assessment Guide for Excipient Makers, Users, and Distributors*.

Both PDA and IPEC Federation believe that presenting a common approach to the legal, regulatory, and related issues concerning excipients is best done as "one voice." Both collaborators see significant potential benefit in leveraging the two organizations' expertise on excipients and drug product manufacture. This technical report will serve as a practical guidance intended for use with existing regulatory and industry standards. The authors expect that the document will enable Manufacturing Authorization Holders of drug product to either set up or benchmark their quality systems, and further establish or continue to collaborate with parties in their excipient supply chain.

This joint PDA-IPEC technical report extends the *PDA Technical Report No. 54* series and provides guidance on risk assessments for excipients by presenting a model risk assessment, guidance on key elements, and a collection of actual examples from excipient users in the pharmaceutical industry.

About Parenteral Drug Association (PDA)

The Parenteral Drug Association (PDA) is the leading global provider of science, technology, and regulatory information and education for the pharmaceutical and biopharmaceutical community. Founded in 1946 as a nonprofit organization, PDA is committed to developing scientifically sound, practical technical information and resources to advance manufacturing science and regulation. Through the expertise of more than 10,000 members worldwide, PDA promotes the exchange of rapidly evolving information on science, technology, and regulations concerning high-quality pharmaceutical production to better serve patients.

About IPEC Federation

The IPEC Federation (IPEC) is a global organization that promotes quality in pharmaceutical excipients. The IPEC Federation represents five regional International Pharmaceutical Excipient Councils (IPECs) — IPEC-Americas, IPEC Europe, IPEC Japan, IPEC China, and IPEC India — and provides a unified voice to promote the best use of excipients in medicines as a means of improving patient treatment and safety. IPEC's objectives are to contribute to the development and harmonization of international excipient standards, the introduction of useful new excipients to the marketplace, and the development of good manufacturing and good distribution practices for excipients.

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1.0 Introduction

Excipients are integrated into the drug product formulation for various reasons: to aid the drug product formulation, to serve as functional or nonfunctional ingredients, or to enable the drug to be released from the drug product in a particular, desired manner. The rationale for their use extends beyond the point of medical use to include maintenance of the safety or function of the drug product, even during storage. While good manufacturing practices (GMP) regulations clearly outline what is required for active pharmaceutical ingredients, only a basic framework exists for ascertaining appropriate GMP for excipients through risk assessment.

In recognition of the criticality of excipients in the production of drug products, the European Commission (EC) Falsified Medicines Directive (2011/62/EU) revised Article 46(f) of Directive 2001/83 to require manufacturing authorization holders to verify that any excipients used in a product are made according to appropriate GMP standards ***(1).*** The Falsified Medicines Directive also committed the EC to publishing *Guidelines on the formalised risk assessment for ascertaining the appropriate GMP for excipients of medicinal products for human use* (EC Guidelines), which were published on March 19, 2015 ***(2)***. In 2018, PIC/S incorporated the same provisions for formalized risk assessment into a publication of the same name, extending the provisions to have global applicability ***(3).*** In the U.S., the 2012 Food and Drug Administration Safety and Innovation Act redefined "current good manufacturing practices" to include "the implementation of oversight and controls over the manufacture of drugs to ensure quality, including managing the risk of and establishing the safety of raw materials [and] materials used in the manufacturing of drugs," in addition to the final drug product ***(4).*** This change also emphasizes the criticality of excipients.

Although the accountability for risk assessment rests entirely with the manufacturing authorization holder, all parties in the supply chain are required to support the process by providing relevant information (e.g., Excipient Information Packages ***(5)***) to the manufacturing authorization holder. The risk management process is applicable throughout the excipient's lifecycle, that is, from selection through termination of use, and remains the responsibility of the manufacturing authorization holder at each stage.

Over the course of an excipient's lifecycle, there will be a broad array of drivers for risk assessment and reassessment. Those drivers may include, among others, product deviations, market recalls, changes in excipient use, supply chain amendments, and adverse trends in quality. The manufacturing authorization holder is responsible for providing evidence that the risk management approach adopted is commensurate with the level of risk to ensure the safety, purity and other quality characteristics of the excipient in use.

To comply with the EC Guidelines and PIC/S publication, each excipient used in a drug product must be assessed for the risks that it poses to the product's quality, safety, and purity. From these risks, the manufacturing authorization holder must determine the quality standards required for the manufacture of the excipient. The manufacturing authorization holder should communicate these quality standards to the excipient supplier and seek the supplier's agreement. Any confirmed gaps in quality standards determined by the assessment must be appropriately mitigated, which may require the manufacturing authorization holder to take action on its own or through collaboration with the supplier. To ensure that risk assessments remain effective during the excipient's lifecycle, the manufacturing authorization holder must incorporate a periodic review into the excipient risk assessment process.

Figure 1.0-1 illustrates this overall, iterative risk management process as it is used in supply chain management ***(6).***

1.1 Purpose

This report is based on the general principles of quality risk management (QRM) as outlined in ICH Quality Guideline *Q9 Quality Risk Management* ***(7).*** It has been developed to provide additional guidance on the excipient risk assessment process required by the EC Guidelines and incorporated into the PIC/S publication. In addition, TR 54-6 presents a model risk assessment and real-world examples

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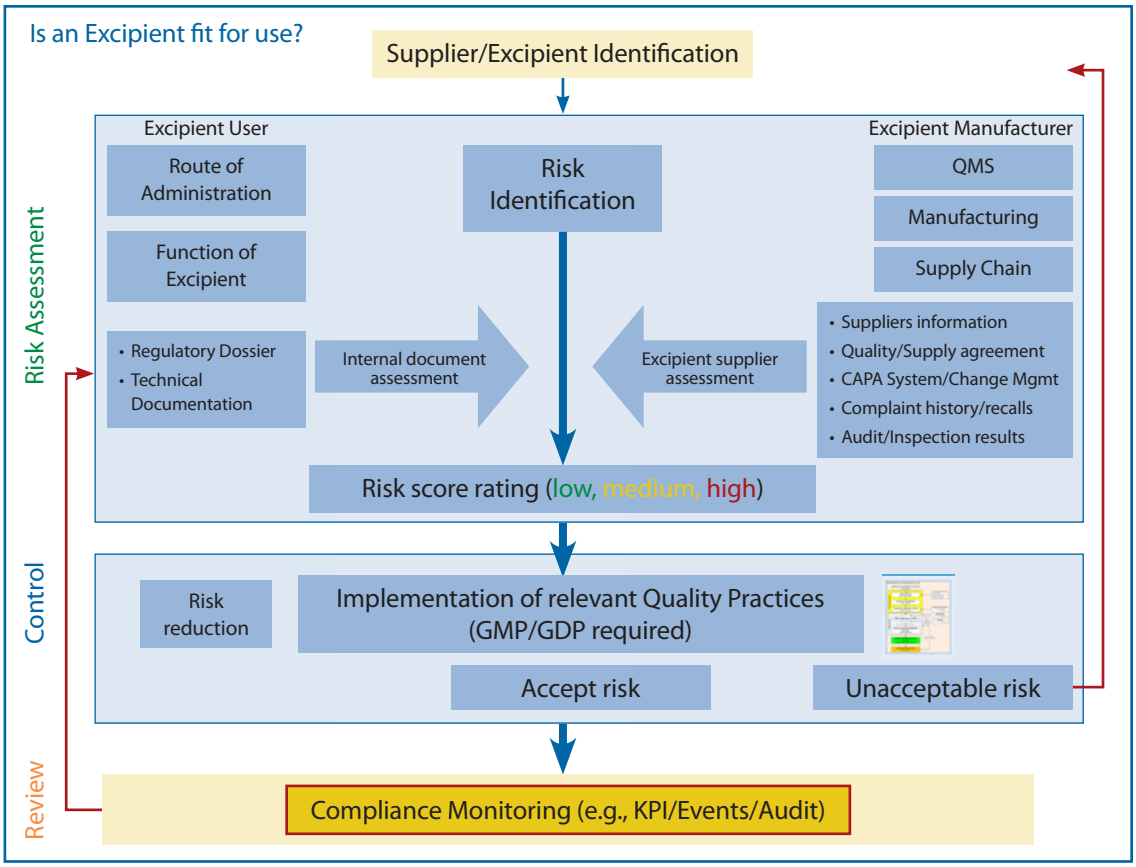

Figure 1.0-1 Identifying Whether an Excipient is Fit for Use (reprinted with permission)

from the pharmaceutical industry applicable to excipient use for all dosage forms to illustrate risk assessment as an excipient knowledge management solution. In general, the model presented in this report is based on the risk ranking and filtering method as referenced in ICH Q9. It also addresses pertinent factors such as end-to-end supply chain, control strategy, and benefits and challenges, among others. TR 54-6 will also serve as a practical guidance intended for use in conjunction with existing regulatory and industry standards; however, it does not replace or alter existing statutory and regulatory requirements for assessing excipients for use in medicinal drug products.

Despite the clarity and guidance offered by the EC Guidelines, risk assessors for manufacturing authorization holders still face some challenges in implementing and benchmarking the excipient risk assessment process for two main reasons. First, quality systems applied during the manufacture of materials used as excipients are diverse and generally based on the material's intended use, which may not be primarily as a pharmaceutical excipient. Therefore, adjusting an excipient manufacturer's or supplier's quality systems to meet the pharmaceutical quality system outlined in the guidelines could prove challenging. Second, an excipient may be used in different dosage forms by manufacturing authorization holders. One manufacturing authorization holder may use a single excipient in both tablets and injectables, which may result in different quality requirements applying to a single excipient from the same supplier. This also may lead to conflicting requirements for GMP.

1.2 Scope

The information in TR 54-6 applies to all excipients used in drug products for human use at all stages of the product lifecycle. It assumes the excipient will be used for the drug product in the commercial supply phase; however, the manufacturing authorization holder may choose to apply this practical guidance in the clinical phase as well. That said, this technical report refrains from providing guidance to the manufacturing authorization holder regarding at which clinical phase it should be applied.

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2.0 Glossary and Abbreviations

Terms used in this technical report are defined in this section. Where a definition is taken verbatim from another source, the source is cited and included in **Section 9.0 References**. All other definitions are specific to Technical Report 54-6. Some definitions are based on those in the IPEC *General Glossary of Terms and Acronyms* ***(8)*** but modified by substituting the term "excipient" for such words as "material" and "product" to aid readability and limit the context.

A listing of all glossary terms used in current PDA technical reports can be found on the PDA Member website. The IPEC General Glossary of Terms and Acronyms is available on the IPEC website.

Broker
A broker is an individual who acts as an intermediary. Brokers resell excipients without conducting physical handling of the excipient such as warehousing, transport, and repackaging. A broker may also be referred to as a trader.

Composite
A composite excipient has two or more components and is designed for enhanced functionality.

Critical Quality Attribute (CQA)
A physical, chemical, biological, or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure the desired product quality (TR54, TR54-4) ***(9, 10).***

Distribution
The division and movement of excipients from the premises of the manufacturer of such excipients, or from another central point, to the end user thereof or to an intermediate point.

Distributor
An entity engaged in distribution that takes legal ownership of the excipient. For purposes of this technical report, the term distributor does not include the original excipient manufacturer.

Drug Product
A finished dosage form (e.g. tablet, capsule, or solution) that contains a drug substance, generally, but not necessarily, in association with one or more other ingredients. The term "drug product" is commonly used by the US Food and Drug Administration. European texts use the equivalent term "medicinal product," which is defined as a substance or combination of substances that is intended to treat, prevent or diagnose a disease, or to restore, correct or modify physiological functions by exerting a pharmacological, immunological or metabolic action (EMA) ***(11).***

EXCiPACT™
A nonprofit organization that manages a certification scheme available to pharmaceutical excipient manufacturers and distributors worldwide, www.excipact.org.

Excipient
Any constituent of a medicinal product other than the active substance and the packaging material (EC) ***(1).*** A substance other than an active pharmaceutical ingredient which has been appropriately evaluated for safety and is intentionally included in a drug delivery system (IPEC).

End User
Organization using the excipient in the manufacture of a drug product.

Haulier/Hauler
Logistics company involved in the movement of an excipient from one location to another.

Original Manufacturer
Person or company manufacturing a material to the stage at which it is designated as an excipient.

Quality Target Product Profile (QTPP)
A prospective summary of the quality characteristics of a drug product that ideally will be achieved to ensure the desired quality, taking into account safety and efficacy of the drug product (ICH Q8 R2) ***(12).***

Relabeling
Process of putting a new label on the excipient. Traceability to the original manufacturer is only assured if the relabeler provides access to its quality systems.

Repacking/Repackaging
Process of putting the excipient into different packaging than that provided by the original manufacturer. Usually synonymous with relabeling.

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Residual Risk
Risk remaining after risk control measures have been implemented.

Risk
The combination of the probability of occurrence of harm and the severity of that harm (ICH Q9) *(7)*.

Risk Analysis
The estimation of the risk associated with the identified hazards.

Risk Assessment
A systematic process of organizing information to support a risk decision to be made within a risk management process. It consists of the identification of hazards and the analysis and evaluation of risks associated with exposure of those hazards.

Risk Priority Number (RPN)
A quantitative measure used when assessing the level of risk.

Shipping Agent
Designated person or agency held responsible for handling shipments, cargo, and the general interests of its customers at ports and harbors worldwide on behalf of ship owners, managers, and charterers. Shipping agents often assist with customs and other clearance routines during import and export processes. They may also transport the excipient and/or hold it in their warehouse.

Storage
The storing of excipients up to the point of use.

Supply Chain
All steps in the entire chain of distribution starting from the point at which an excipient is transferred outside the control of the original manufacturer's material management system downstream to the final user of the excipient (IPEC). It is a system of organizations, people, technology, activities, information, and resources involved in moving an excipient from the original manufacturer to the end user.

Supplier
Person or company providing excipients on request. Suppliers may be distributors or traders, among others. Supplier is a generic term that can be used to describe any of the actors in the supply chain but, for the purposes of this technical report, it does not include the manufacturer.

Trader
See Broker.

Warehousing Company
Organization providing a warehouse that may be used to store excipients and other materials.

2.1 Abbreviations

EC European Commission
CMO Contract Manufacturing Organization
CQA Critical Quality Attributes
GMP Good Manufacturing Practices
QMS Quality Management System
QRM Quality Risk Management

3.0 Overall Strategy – A Holistic View

TR 54-6 provides guidance on quality risk management principles that can be used to assess the risks to the quality, safety, and function of an excipient in a drug product. The proposed generic model allows the user to answer the risk question "Is the excipient fit for use in the drug product" through the use of a formalized risk assessment. The model, as described in **Section 6.0** and illustrated in **Figure 6.0-1**, describes the steps involved in identifying whether an excipient is fit for use considering an assessment of its use and function for the excipient user and an assessment of the entire supply chain.

In the model, the potential hazards from the excipient itself, as well as the supply chain, are ranked and evaluated using a risk ranking and filtering approach to evaluate the overall risk of harm to the drug product and, through that, to patient safety.

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Based on the results of the risk assessment, a pharmaceutical manufacturer can determine what level of GMP will be required. The guides and standards most widely accepted for excipient GMPs are:

- EXCiPACT™: International Pharmaceutical Excipients Certification Standards for Pharmaceutical Excipient Suppliers ***(13)***
- NSF/IPEC/ANSI 363: Good Manufacturing Practices (GMP) for Pharmaceutical Excipients ***(14)***
- IPEC/PQG: The Joint Good Manufacturing Practices Guide for Pharmaceutical Excipients ***(15)***
- USP: USP <1078> Good Manufacturing Practices for Bulk Pharmaceutical Excipients and <1059> Excipient Performance ***(16, 17)***
- Japan: Self-imposed Standards of Good Manufacturing Practices for Pharmaceutical Excipients ***(18)***

The pharmaceutical manufacturer may determine that some additional technical considerations beyond those in established GMP guides and standards may be needed. As part of the risk assessment process, the manufacturer or manufacturing authorization holder of the drug product should engage the excipient manufacturer and suppliers to obtain any specifics from the excipient manufacturer's own risk assessment process that the excipient manufacturer is willing to share. Additionally, the complete supply chain profile must be considered for each excipient.

Should the risk assessment uncover an issue requiring additional action, a control strategy should be put in place and the affected parties should be notified.

Before being implemented, any actions recommended as a result of the risk assessment must be discussed and agreed upon by the excipient manufacturer and the manufacturer of the drug product or manufacturing authorization holder, in case those actions may impact the supplier. The results of the discussion should be documented in a quality agreement or other document.

A global risk assessment of all excipients used by the drug product manufacturer or manufacturing authorization holder will ensure a standardized assessment of all excipients in all drug products and provide a holistic view of the supply chain. This enables the excipient user to provide the excipient manufacturer, with one voice, a single (worst case) risk assessment. That approach is preferable to having multiple users within a single drug product manufacturer each communicating different assessment results to the excipient manufacturer.

In a global risk assessment, risk review and improvements of the risk assessment might be more difficult than they would be if separate risk assessments were completed for each drug product or by each site using the excipient. The review and approval workflows for a single global risk assessment may require more time and effort. However, this process could be shortened by instituting a global risk assessment that involves local ownership. In this scenario, the owner of each product or manufacturing site would conduct a separate risk assessment, but all risk assessments would be collected in a single document repository. The local owner would maintain responsibility for the data collection, review, and approval workflows, and for ensuring that updated information is included in the global risk assessment repository. The repository would serve as a knowledge base for all excipients, allowing the drug product manufacturer or manufacturing authorization holder to communicate its requirements to the excipient manufacturer with a single voice representing all local owners.

3.1 Risk Areas Beyond Quality

While the focus of the excipient risk assessment is quality, a company also should consider other aspects that can impact the excipient and, with this, product availability for the patient. These include excipient availability; business continuity risk; environmental, socio-economic, environment, health, safety, and geographic risks; and general business data. While consideration of these risks is useful as part of an overall strategy, it is out of scope of this document and is not discussed further.

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4.0 Supply Chain — End to End

Section 4.0 provides examples of supply chains, helps identify the various actors in the supply chain, and describes their typical roles. The subsections herein offer examples of the types of information and documentation that the various actors could provide. Additionally, it discusses some aspects of the potential risks that could be introduced by actors in the supply chain.

Supply chains have become more and more complex over the years. In the past, a manufacturer of pharmaceutical excipients often supplied directly to a drug product manufacturer. Today, in addition to the manufacturer of the excipient, the distributor, broker, and contract manufacturing organization (CMO) each plays a role in the supply chain, and each contributes to the risks to quality in one way or another. Each of these companies also could provide information on the excipient's manufacture, testing, and distribution that might be relevant for the risk assessment.

4.1 Types of Supply Chains and Risks Presented

Many materials used as excipients are not manufactured specifically for pharmaceutical use, but are intended for other applications such as food, cosmetics, or home care. As such, the direct supply of an excipient from the manufacturer to the end user may be unusual. In addition, the quantities of excipients used by the pharmaceutical end user may be only a fraction of the excipient manufacturer's overall volume; in fact, the original manufacturer may not have a business interest in supplying the pharmaceutical industry. In such cases, intermediaries often become involved, procuring larger quantities of the excipient from the original manufacturer, and then repackaging and relabeling it into quantities more suitable for pharmaceutical use. Thus, the supply chain between the excipient manufacturer and the pharmaceutical end user often is complex. The complexity is magnified for excipients that move internationally.

In both international and domestic supply chains, the point at which the legal ownership of the excipient transfers from one organization to another is a critical juncture. Each organization needs to have defined responsibilities for maintaining records and securing the quality and security of the excipient. In complex supply chains, legal ownership may transfer several times, for example from original manufacturer to distributor to repackager to end user.

In all cases, the end user needs to identify the intermediaries in the supply chain in order to properly assess the risks. This may be difficult, as some organizations in the supply chain may not want to disclose their sources, even including the original manufacturer of the excipient.

This **Section 4.1** discusses some of the risk factors inherent to transferring excipients through various configurations of supply chains.

4.1.1 Direct Supply from Manufacturer

Even when an excipient is supplied directly by a manufacturer to an end user, intermediaries may be involved. Logistics operations may be subcontracted to effect movement of the excipients. One or more warehousing companies also may be involved. Because all intermediaries are subcontracted by either the original manufacturer or the end user, they should have good information about how to safely transport the excipient. The supply chain illustrated in **Figure 4.1.1-1** is one of the lowest-risk arrangements due to the small number of intermediaries and the degree of control that can be applied.

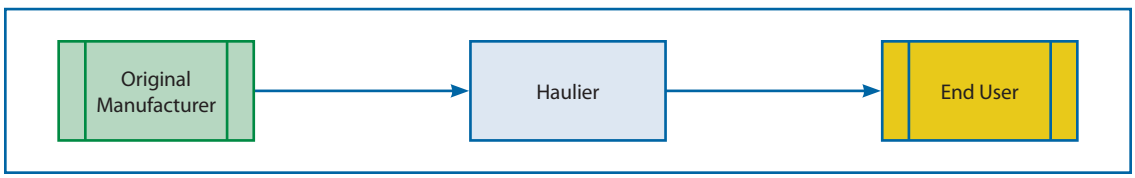

Figure 4.1.1-1 Supply Chain: Direct Supply from Manufacturer

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4.1.2 Supply via a Distributor

When an excipient moves from manufacturer to distributor, the distributor becomes the legal owner. The distributor then transfers ownership to the end user, possibly after repackaging or relabeling the excipient. The distributor may have an agreement with the original manufacturer to market the excipient using the distributor's tradename and literature; in this instance, the distributor and manufacturer likely will have an agreement to ensure the quality of the excipient. Usually in these cases the entire supply chain can be mapped with some certainty. This type of supply chain, depicted in **Figure 4.1.2-1,** presents few more risks than the simple manufacturer-to-end user scenario; however, those risks may be magnified if the distributor engages in repacking or relabeling, as discussed in **Section 4.1.3.**

If the original manufacturer has not established agreements with distributors, it will be much harder to fully trace all parties in the supply chain, increasing the risks. Other parties may repack and/or relabel the excipient. The original manufacturer may change its distributor, hence the process for notification of such changes should be agreed between the parties.[1] These supply chains are more complex to assess, and the attendant risks are higher than in the previous scenarios.

4.1.3 Supply Via Repackager

Repackaging activities in the supply chain, as illustrated in **Figure 4.1.3-1,** magnify the risks to excipient quality. The end user needs to understand the GMP controls applied by the repackager in order to minimize the risks to excipient quality, and, if feasible, needs to specify appropriate measures in supply or quality agreements.

4.1.4 Supply Via Broker

A broker usually connects the manufacturer with the customer. **Figure 4.1.4-1** depicts a possible relationship between a distributor and the end user when a broker is involved.

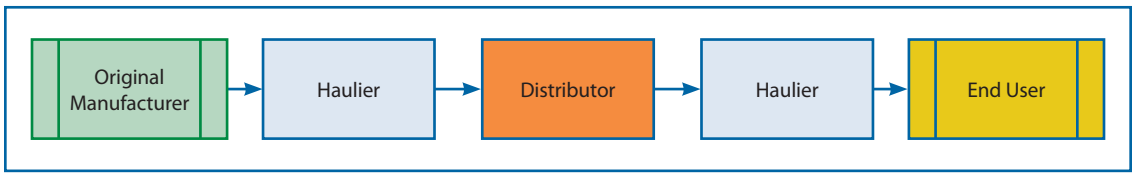

Figure 4.1.2-1 Supply Chain: Supply via Distributors

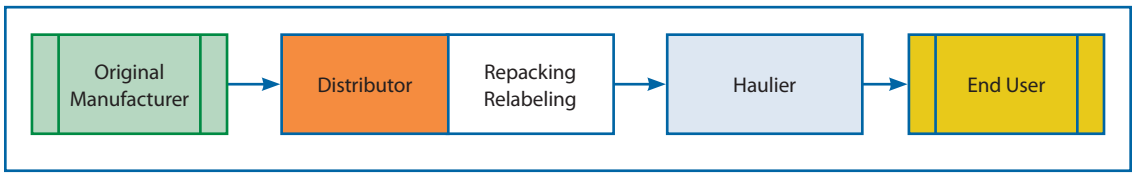

Figure 4.1.3-1 Supply Chain: Supply via Repackager

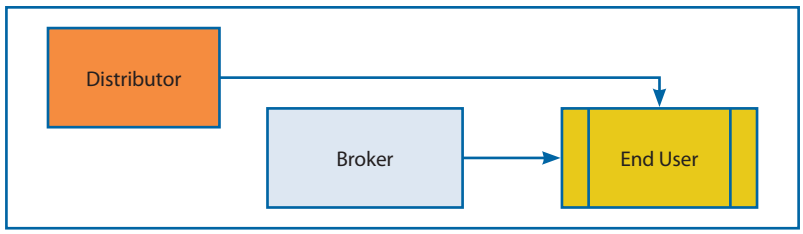

Figure 4.1.4-1 Supply Chain: Supply via Broker

[1] The *IPEC Significant Change Guide for Pharmaceutical Ingredients* provides a uniform approach to the significance of changes involving the manufacture of pharmaceutical excipients and may be a useful resource ***(19).***

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The ability to map the supply chain in this scenario will depend on the degree of transparency provided by the broker. When communications are totally open, the supply chain can be traced, and any risks can be assessed clearly. For instance, if the broker is fully transparent and the supplier is the original manufacturer, the risks may be as low as in the direct shipment scenario described in **Section 4.1.1.** On the other hand, if the broker is less transparent, it may be very difficult to assess the risks, indeed.

4.1.5 Manufacturer Exporting Excipient Directly to End User

An international supply chain such as that illustrated in **Figure 4.1.5-1** usually involves many intermediaries. Unlike previous examples in which the parties are relatively fixed, the parties to an international supply chain, including haulers, shipping agents, and ships, may vary from shipment to shipment. In these situations, quantifying the risks may be very difficult. If the intermediaries are fixed, however, suitable inspection controls can be applied at those points, reducing the uncertainty presented by the prior steps.

4.1.6 Manufacturer and Contract Manufacturing Organizations

When one or more CMOs are involved in the supply chain, it is critically important to clearly identify and understand which party or parties gather information and who owns the risk assessment.

The manufacturing authorization holder is ultimately responsible for the risk assessment. Depending on the quality agreement and relationship between the CMO and manufacturing authorization holder, there generally are three options for conducting the risk assessment:

- The CMO might carry out the entire risk assessment, collecting the information about the use and function of the excipient in the final drug product. That risk assessment would be reviewed and approved by the manufacturing authorization holder.
- The risk assessment might be carried out by the manufacturing authorization holder, incorporating information from the CMO.
- The risk assessment might be carried out by the manufacturing authorization holder based on the risk assessment delivered by the CMO.

In any case, the roles and responsibilities for the CMO and manufacturing authorization holder must be clearly defined, including what actions will be taken if the risk assessments completed by the CMO and manufacturing authorization holder result in different outcomes.

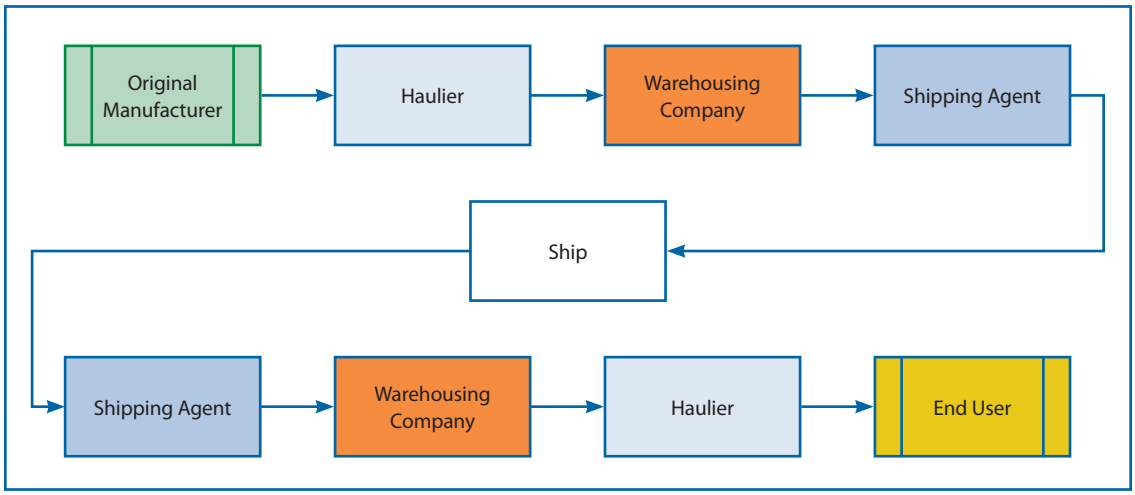

Figure 4.1.5-1 Supply Chain: Manufacturer Exporting Excipient Directly to End User

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5.0 Different Roles in the Supply Chain and Information-Gathering

To successfully complete the formalized risk assessment process, the information and evidence obtained must support conclusions on risks, identify gaps in GMP, and determine control strategies. The risk assessment process is applicable throughout the lifecycle of the excipient—from selection to termination of use—and remains the responsibility of the manufacturing authorization holder. This key task, however, requires collaboration from all other parties in the supply chain. **Table 5.0-1** delineates the roles in the supply chain, examples of their areas of responsibility, the information for which they are accountable, and the types of documents they may be expected to supply.

Table 5.0-1 Roles in the Supply Chain and Examples of Their Responsibilities

Supply Chain Role	Areas of Accountability	Information Accountable	Documents Typically Provided
Subcontractor of the Excipient Manufacturer	– Sourcing raw materials – Manufacturing – Testing – Packaging and labeling – Transporting	– Manufacture – Testing certificates – Compliance certificates – Quality standards in use	– Manufacturing flow chart – Certificate of analysis – ISO, GMP, or EXCiPACT™ certificate – Supply chain map – Transport and storage conditions
Manufacturer of the Excipient	– Sourcing raw materials – Manufacturing under appropriate standards – Testing – Packaging and labeling – Transporting	– Manufacture – Testing certificates – Compliance certificates – Quality standards in use	– Manufacturing flow chart – Certificate of analysis – ISO, GMP, or EXCiPACT™ certificate – Supply chain map – Transport and storage conditions – Excipient information package (EIP) or comparable information package – Regulatory declaration statement – Technical expert support – Excipient master files (e.g., U.S. DMFs)
Broker	– Flow of information	– Supply chain overview – Access to suppliers	– Certificate of analysis – Supply chain map
Supplier or Distributor	– Transporting – Warehousing – Repackaging – Testing	– Transport overview – Storage conditions – Direct link to manufacturer/supplier – Packing and labeling – Testing certificates – Compliance certificates – Quality standards in use	– Access to excipient information package (EIP) or comparable information package – Regulatory declaration statement – Certificate of analysis – ISO, GMP, or EXCiPACT™ certificate – Supply chain map – Transport and storage conditions – Technical expert support
CMO or Manufacturer of Drug Product	– Sourcing raw materials – Manufacturing drug product – Testing of excipient – Packaging and labeling – Transporting – Conducting excipient risk assessment (Ultimate accountability rests with manufacturing authorization holder)	– Excipient information – Supply chain oversight – Formulation information – Special requirements of excipient	– Excipient risk assessment – Supply chain map – Audit of excipient supplier and manufacturer – Quality oversight of supplier and manufacturer – Regulatory filing of excipient – Toxicological data – Excipient quality and performance quality trends of excipient – Formulation information – Specification of excipient – Control strategy
Marketing Authorization Holder	– Submission of dossier – Defining route of administration and function of excipient – Excipient risk assessment	– Toxicological data – Maximum daily dose – Excipient use and function	– Technical data – Regulatory file – Manufacturing process flow – Supply chain overview – Manufacturer and supplier audit documents – Oversight of outsourced activities (e.g., risk assessment, compliance history)

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Information to support the excipient risk assessment process can be obtained using a variety of avenues:

- Meetings with manufacturers and suppliers
- Obligations placed in quality agreements or equivalent documents
- Chemistry, manufacturing and controls (CMC) documentation
- Excipient manufacturer customer portals
- Internet
- Manufacturing authorization holder supply chain management and quality oversight functions
- Health Authority portals
- Quality and technical collaborations between excipient manufacturer and end user

That said, when collaboration between the manufacturing authorization holder and other parties in the supply chain is limited, the manufacturing authorization holder may need to consider alternative means for obtaining information. **Table 5.0-2** lists some common scenarios and potential resolutions.

Table 5.0-2 Common Scenarios in Information-Gathering

Potential Issues	Manufacturing
Unable to determine all participants in the supply chain for commercial reasons, e.g., a distributor enforces supply chain opaqueness for competitive reasons	When distributors are part of the supply chain, this is a common scenario. The manufacturing authorization holder should identify the supply chain risks and, where appropriate, transfer responsibility for supply chain oversight to the distributor. – Allow the distributor to redact information in responses to protect their business advantage – Use the quality agreement with suppliers/distributors to assign accountability and supply chain oversight – Use onsite audit program to challenge and confirm information
Unable to obtain appropriate regulatory declaration statements (e.g., residual solvent statement) for an excipient due to the lack of technical understanding or expertise of a participant in the supply chain	– Support the excipient manufacturer/supplier with in-house technical support, if available – Provide the manufacturer/supplier with declaration templates, if available, for manufacturer/supplier to complete – Use the manufacturing authorization holder's internal technical team to assess and/or test material in lieu of supplier regulatory declaration
Unable to obtain all information required to support the risk assessment because the excipient manufacturer's quality standards do not meet the excipient user's requirements	– Determine if the excipient manufacturer is performing additional activities to facilitate alignment of quality standards – Reassess to identify gaps that exist and mitigate through internal or external action, e.g., increase quality oversight, change specifications – If the residual risks identified are unacceptable, change the excipient supply chain
Insufficient provisions in place (e.g., lack quality agreements or unable to agree on a confidentiality agreement) to support information exchange with manufacturing authorization holder	– Review obligations agreed with parties in the supply chain – Put in place an appropriate agreement to support this process – Review manufacturer audit reports and supply management files to determine whether the assessor for the manufacturing authorization holder already possesses required information – Explore other avenues of obtaining information, e.g., manufacturer's standard customer information portals, public domain information
Insufficient or no collaboration between key participants and manufacturing authorization holder in the supply chain (e.g., no leverage due to low volume purchase, stressed or non-existent commercial relationships)	– Use distributor relationship to leverage information transfer – Identify alternative sources for obtaining support and information – Use already-established agreements or put in place appropriate agreement to obligate supply chain party support – Review manufacturer/supplier audit reports/questionnaire/excipient file (e.g., EIP) and supply management files to determine whether the assessor for the manufacturing authorization holder already possesses required information
Unable to support the excipient risk assessment process due to limited oversight of outsourced activity (e.g., unable to secure an audit)	– Ensure adequate resources are allocated to ensure appropriate prioritization of manufacturer/supplier management activities – Use third-party audit program (e.g., EXCiPACT™)

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6.0 A Model for Quality Risk Management for Excipients

Experts on the TR 54-6 Task Force conducted an informal survey examining industry practices relating to quality risk management for excipients. Based on analysis and discussion, they developed a model combining best practices. In addition, the Task Force assessed real-world company models after several manufacturing authorization holders shared examples of actual risk assessment processes. See **Annex 1** (**Section 10.0).**

In general, the proposed QRM model for excipients is based on the risk ranking and filtering method. Risk ranking and filtering is a tool for comparing and ordering risks. Risk ranking of complex systems typically involves evaluation of multiple diverse quantitative and qualitative factors for each risk. The tool involves breaking down the basic risk question "Is the excipient fit for use?" into as many components as needed to capture factors involved in the risk.

The two main factors in the model are the intrinsic excipient risks and the supply chain risks. These factors are combined into a single relative risk score that can then be used for ranking risks. Filters, in the form of weighting factors or cut-offs for risk scores, are used to support risk acceptance or mitigation. **Figure 6.0-1** illustrates the steps in the process. In other sections below, this TR highlights and discusses portions of this model.

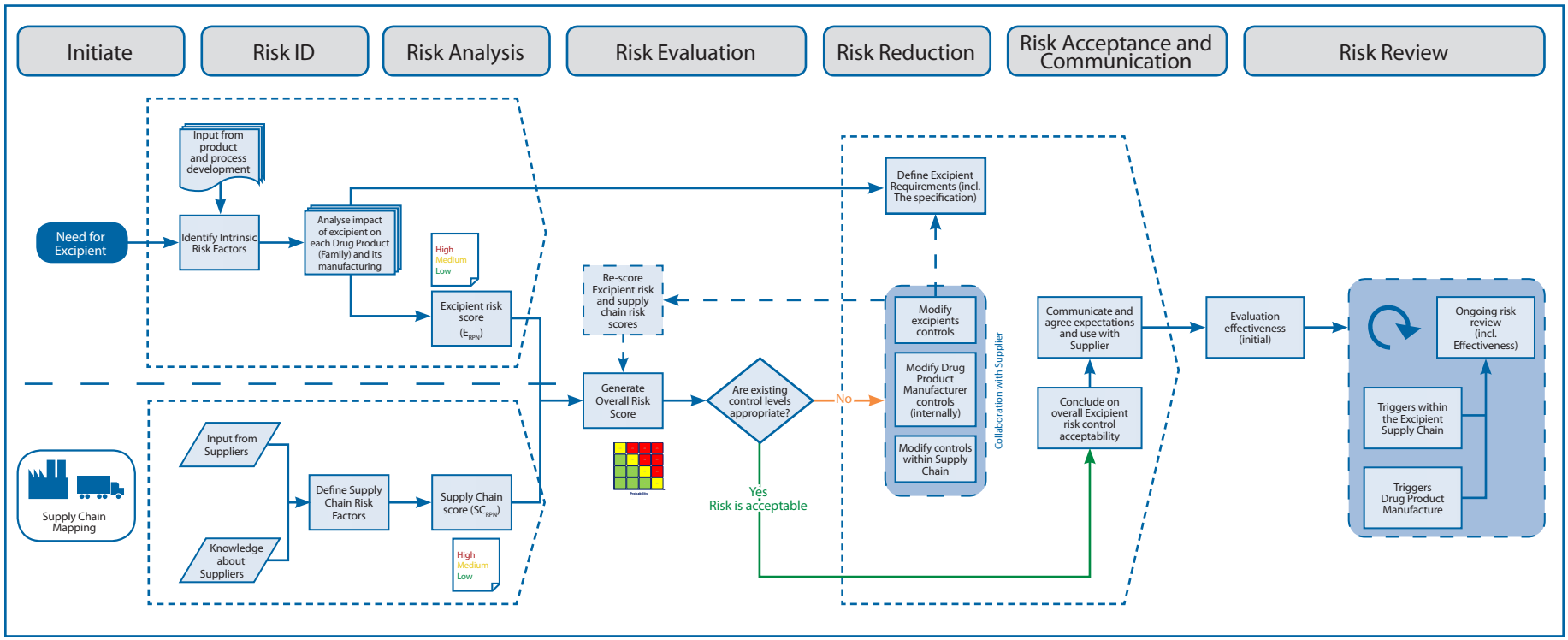

Figure 6.0-1 The Complete Generic Risk Model for Excipients

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6.1 Initiate Risk Management Activities and Identify Assessment Team

The risk assessment process begins with the identification of a need to qualify an excipient or to review an existing excipient risk assessment (e.g., at new product introduction, as a result of obtaining a second source, being notified by the supplier of major changes, or one of the numerous other drivers discussed below in **Section 7.1).**

A successful excipient risk assessment program requires a holistic approach encompassing knowledge of the excipient's use in the pharmaceutical dosage form as well as knowledge of the excipient's manufacturer and supply chain. The assessment team should include individuals from various functional areas, including subject matter experts (SME) from the drug product manufacturer who understand the excipient's use in the drug product, as well as individuals from the functions involved in oversight of the supply chain.

The risk assessment process might include participants with defined roles and responsibilities in the functional QRM areas and others, such as:

- QRM—risk assessment facilitator knowledgeable about the methodology and tools
- QRM experts—area or function leads for QRM or the QRM decision-makers
- Quality Assurance—focused on use and function
- Quality—focused on oversight activities related to supply chain (e.g., excipient manufacturer, supplier, transport)
- Qualified/responsible person
- Material sciences
- Quality control
- Regulatory affairs
- Technical/operations
- Sterility assurance
- Manufacturer or supplier auditing
- Procurement
- Outsourcing operations—oversight activities
- R&D
- Engineering
- Nonclinical statistics
- Legal
- Manufacturing
- Medical/clinical
- Safety and pharmacology/toxicology

Having a multidisciplinary team of individuals working collaboratively to gather and review data and assess the potential for risk is critical to ensuring that all risks are considered holistically. The excipient risk management process should be built into a company's existing processes and quality management system (QMS).

6.2 Identify Intrinsic Risk Factors

In identifying the excipient's intrinsic risk factors, numerous factors should be considered.

One primary source is input from product and process development. The Quality Target Product Profile of the drug product(s) in which the excipient is to be used provides requirements for the drug product, and this forms the basis of information about the use and function of the excipient. The manufacturing authorization holder assesses the drug product requirements to develop the drug product critical quality attributes (CQA) and to establish the foundation for drug product design and development. During the drug product's design and development, scientific studies are conducted and information is collected concerning the excipient, particularly about how it may interact with the product's other components. Drug product development and process design studies, commercial manufacturing, and associated risk assessments for this product and any other products in which the excipient has been used, provide historical information about the excipient that can be used to assess its impact on the targeted drug product.

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In addition, examples of parameters that can be considered as excipient attributes include:

- Characteristics of excipient needed for use in the drug product
- Manufacturer's excipient specifications
- Factors learned from the manufacture of the excipients
- Existence of multiple and or global manufacturers
- Lot-to-lot variability
- Risk factors listed in the EC Guidelines of March 19, 2015, such as:
  - Transmissible spongiform encephalopathy (TSE)
  - Potential for viral contamination
  - Potential for microbiological or endotoxin/pyrogen contamination
  - Potential, in general, for any impurity originating from the raw materials, e.g., aflatoxins or pesticides, or generated as part of the process and carried over, e.g., residual solvents and catalysts
  - Sterility assurance for excipients claimed to be sterile
  - Potential for any impurities carried over from other processes, in the absence of dedicated equipment and/or facilities
  - Environmental control and storage/transportation conditions, including cold chain management if appropriate
  - Excipient stability
  - Packaging integrity evidence

6.3 Excipient Risk Analysis

The excipient risk analysis can be carried out in two steps. First, an initial excipient impact assessment can be conducted to provide input to the internal specification of the excipients, to identify intrinsic risk factors relating to the excipient's impact on the drug product. This assessment is used in the second step—the subsequent risk-scoring of the excipients. **Figure 6.3-1** focuses in on one section of **Figure 6.0-1** to illustrate the initiation of the risk assessment when an excipient is needed for a product.

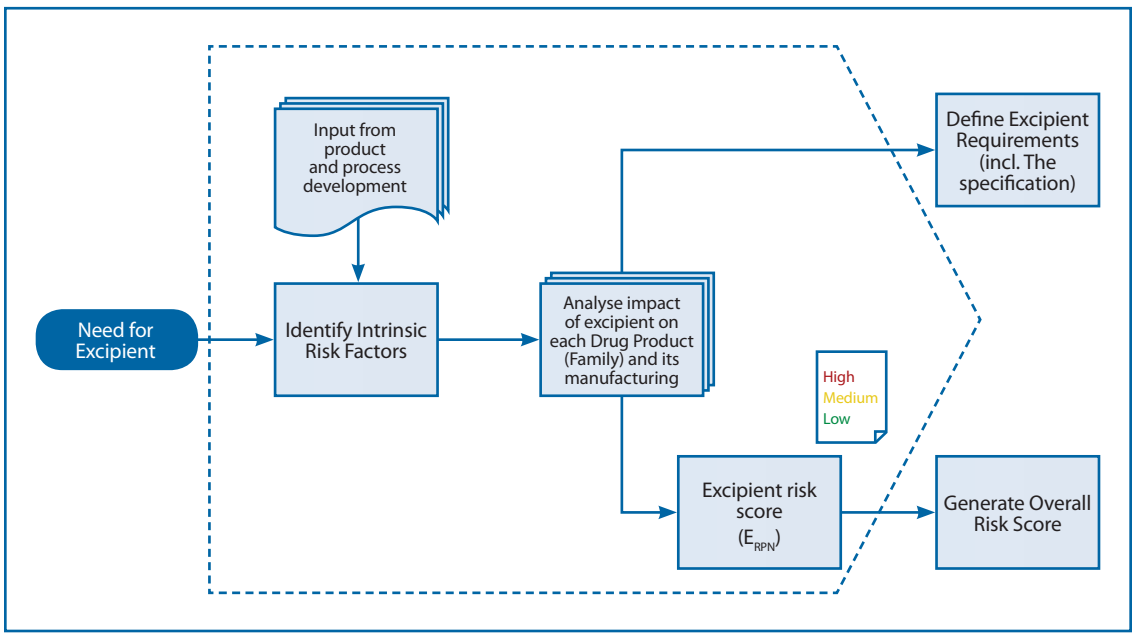

Figure 6.3-1 Steps in Excipient Risk Identification and Analysis

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6.3.1 Impact of Excipient on Each Type of Drug Product

After identification as described in **Section 6.2,** the excipient attributes are assessed for correlation to the CQAs of the drug product in which the excipient is to be used. The rationale of the impact must be explained. This step must be taken for each drug product or drug product family. Then, the results can be consolidated across products.

Initially, all potential risk factors (hazards) are identified and documented as excipient attributes. Then, CQAs are correlated to the pharmaceutical dosage form, as follows:

+ Known Impact (documented effect)
- No Impact (no indication of impact)
? Impact cannot be ruled out, more information may be needed

The information can be compiled in a table, such as that shown in **Figure 6.3.1-1.** A column marked "Compliance" is included near the CQAs so that compliance with, for instance, pharmacopeial specifications, can be ensured as well. This column is useful because, in some cases, an attribute may be part of a specification, e.g., a pharmacopeial specification, that the company will claim adherence to, regardless of the actual impact on the drug product CQAs. This column allows the assessors to force such excipient attributes further into the assessment and into the excipient specification.

	Assess impact of Excipient on CQAs					Assess failure of Excipient		Propose specification	Spec. input
	Drug Product CQA	Drug Product CQA	Drug Product CQA	Drug Product CQA	Compliance	Failure options	Describe known effect of failure	Assess specification interval that will prevent failure	
Excipient attribute	-	-	-	-	+	Text	Text	Limit or interval	
Excipient attribute	-	-	-	-	-	N/A	N/A	N/A	
Excipient attribute	+	-	-	?	+	Text	Text	Limit or interval	
Excipient attribute	-	+	-	-	-	Text	Text	Limit or interval	

Figure 6.3.1-1 Illustration of an Excipient Impact Assessment to Provide Specification Input

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Only excipient attributes with a correlation marked "+" are brought forward as needed into the specification for further analysis. The attributes marked "?" should be further investigated to obtain the data necessary to determine whether there is potential impact on the drug product. If further data cannot be gathered, "?" will be treated as a "+," taking a worst-case approach.

Excipient attributes and impact assessments are used to carry out a qualitative failure analysis in which the failure options and effect of failure (consequences) are described. This analysis is based on the knowledge gathered during the development phase of the drug product or from other literature and knowledge.

From the failure analysis, the excipient user gains understanding of potential and allowed variation in the attributes. With this knowledge, specification limits for each material attribute are proposed to mitigate the risk of impact. In this way, the excipient impact assessment is used to justify the excipient specification.

An impact assessment is conducted for each drug product in which the excipient is used. If done retrospectively, this impact assessment leads to an evaluation of the existing specification. If done prospectively, the impact assessment provides input to one or more specifications consistent with drug product needs. In most cases, it may be advantageous to have a single specification that applies to the excipient's use in all drug products.

6.3.2 Excipient Risk Profile based on Use and Intrinsic Factors

The second part of the excipient analysis is creating a risk profile (or ranking) using a question-based risk ranking and filtering system.

At a minimum, the following factors related to the excipient and its use should be considered in developing the risk score, as explained further in **Table 6.3.2-1:**

- Excipient attributes and relevant specification
- Functionality of the excipient in formulation
- Intended patient intake (consider dose regimen, frequency of dose, strength of formulation)
- Route of administration
- Excipient origin and potential for contamination
- Excipient complexity (composite)
- Prior knowledge of and experience with excipient (e.g., known quality defects)
- Standard packaging size of the excipient (consider use in manufacture of drug product)

Table 6.3.2-1 describes different scenarios expressing a LOW, MEDIUM, or HIGH risk for each of the risk factors **(RFE)** listed. This table can be used as a starting point for consideration of (potential) risk factors, then developed and improved over time to reflect the particular excipient and its use in the final drug, as well as the user's experience with the model. While it includes the risk factors required by the EC Guidelines, other relevant risk factors can be added.

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Table 6.3.2-1 Excipient Risk Calculation Tool

Risk Factor Excipient (RF E)	Low (1)	Medium (3)	High (5)
RF E_1: Functionality of Excipient in Formulation	Diluent, coloring agent flavor, sweetening agent, identity through printing with ink, emollient, tonicity agent	Thickener, coating agent, former, compression aid, lubricant, glidant and/or anticaking agent, humectant, suspending/dispersing agent, buffer, adhesive, penetrant, disintegrant, binder, capsule shell, plasticizers, pH modifiers, chelating and/or complexing agents, antioxidants, suppository base, suspending and/or viscosity-increasing agent, stiffening agent, propellant	Antimicrobial preservative, vehicle, release modifier, wetting and/or solubilizing agent, pH modifier, pharmaceutical water
RF E_2: Intended Patient Intake (Consider dose regimen, frequency of dose, amount of excipient in dose form); consult the Inactive Ingredients Database ***(19)***	Expected maximum daily intake does not exceed known maximum dose of excipient in a drug product	N.A.	Expected maximum daily intake exceeds known maximum dose of excipient in a drug product
RF E_3: Route of Administration (Risk to the patient)	Topical, oral, vaginal, rectal	Pulmonary/inhalation	Parenteral, ophthalmic, for open wounds
RF E_4: Potential for Contamination	Excipient is well-defined, not of animal origin; manufacturing process is well-controlled, resulting in low or no potential for contamination by: – TSE/BSE – Viral contamination – Microbiological organisms, endotoxins, or pyrogens – Impurities resulting from the manufacturing process, e.g., residual solvents and catalysts Impurities from the raw materials, e.g., aflatoxins or pesticides	N.A.	Animal origin or circumstances indicating a heightened potential for contamination by virus, impurities, or other unwanted adulterants
RF E_5: Excipient Complexity (Is it a composite or individual excipient?)	Excipient is not a composite	Excipient is a composite and each component of the composite is well characterized and supported by technical data	Excipient is a composite and components are not formally managed or traced
RF E_6: Prior Knowledge or Experience with Excipient (e.g., known quality defects)	Material does not cause challenges; prior experience with the excipient and low or no history of defects	Prior knowledge or experience with the excipient is minimal	History of defects and challenges, or no prior knowledge of or experience with the excipient
RF E_7: Packaging Size (Consider internal use in manufacture of product)	Single-use, ready to be used in manufacturing	Excipient must be divided or repackaged for manufacturing, but transfer of excipient into alternative primary packaging does not introduce additional risks	Excipient must be divided or repackaged in a highly controlled environment with high precision before use in manufacturing
RF E_x: Add according to specific needs for the excipient in final drug product	To be defined	To be defined	To be defined

Each selection will result in a value of 1 (LOW), 3 (MEDIUM), or 5 (HIGH) for risk.

The risk score for the excipient (**E_{RPN}**) is calculated as the sum of the risk scores for each of the seven identified risk factors (**RF E**)[2], divided by the number of risk factors. The resulting E_{RPN} will be in the range of 1 (**Low**) to 5 (**High**).

$$\textit{Overall Excipient Risk Score } (E_{RPN}) = \frac{RF\,E_1 + RF\,E_2 + RF\,E_3 + RF\,E_4 + RF\,E_5 + RF\,E_6 + RF\,E_7}{7}$$

[2] Users would adjust the calculation as necessary to reflect additional risk factors that they have identified for consideration or to weight the individual risk factors based on the company's analysis and risk tolerance.

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The resulting risk score (**E_{RPN}**), along with a similar overall score for the supply chain, will be used to evaluate the suitability and acceptability of the current excipient setup.

6.4 Supply Chain Risk Analysis

A supply chain risk score (**SC_{RPN}**) is created using a risk ranking system similar to that used for evaluating the excipient risk score. The ranking should consider input from the supplier, including knowledge from questionnaires, as well as knowledge gained through previous experience, audits, visits, and media coverage, as illustrated in **Figure 6.4-1.**

At a minimum, the following risk factors related to the supply chain (**RF S**) should be considered:

- Supply chain complexity—supplier, broker, manufacturer relationships, opportunity for fraud (See **Section 4.1.)**
- Prior knowledge of the supply chain—capability
- Organizational volatility
- Excipient manufacturer performance history—oversight knowledge such as customer complaints, changes, audits, trustworthiness, tailgate samples
- Packaging suitability—ability to protect excipient from fraud, moisture, heat, and similar elements
- QMS standard and certification

Table 6.4-1 describes different scenarios expressing a LOW, MEDIUM, or HIGH risk for each of the risk factors **(RF S)** listed. This table can be used as a starting point for consideration of (potential) risk factors, then developed and improved over time to reflect both the particular excipient and its use in the final drug, and the user's experience with the model. While it includes the risk factors required by the EC Guidelines, other relevant risk factors can be added.

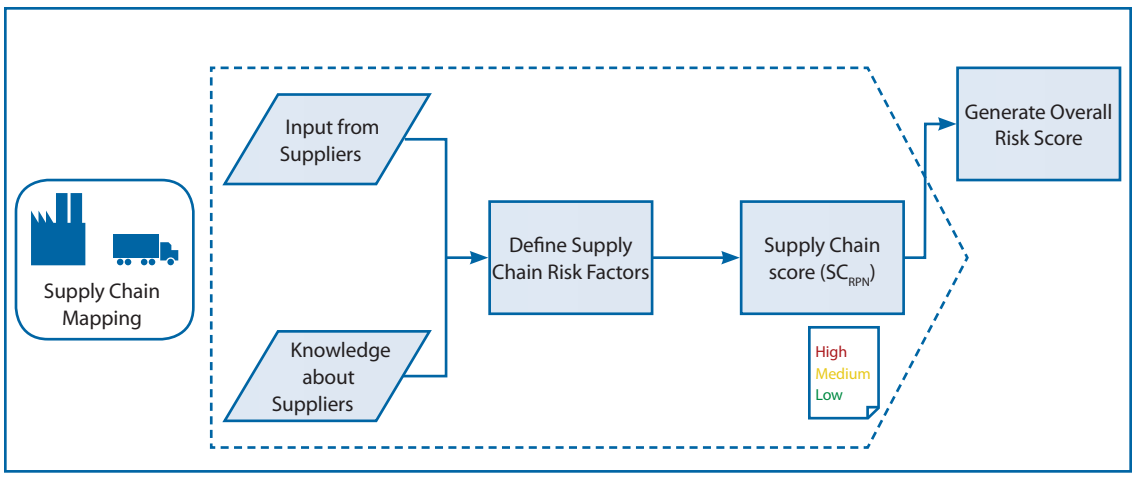

Figure 6.4-1 Steps in the Supply Chain Risk Analysis

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Table 6.4-1 Supply Chain Risk Calculation Tool

Risk Factor Supply Chain (RF S)	Low (1)	Medium (3)	High (5)
RF S_1: Supply Chain Complexity (Consider supplier, broker, manufacturer relationships, considering the opportunity for fraud)	Purchased directly from manufacturer–low probability of fraud; sourced from a region with known and transparent controls on counterfeiting and security (i.e., US, EU); excipient not repackaged or manipulated after manufacturing process; transport from supplier is through qualified carrier	A least one broker, distributor, or intermediate supplier, but there is transparency to the original manufacturer and location; supplier re-manipulates (i.e., repackages, relabels) under established and understood GMP procedures	Supply chain is complex with several manipulations or distribution channels; repackaging and/or manipulation occurs after the manufacturing process; region of security concern and potential for fraud
RF S_2: Prior Knowledge of the Manufacturer (Consider capability)	Prior experience with manufacturer; critical attributes monitored and controlled; no excipient nonconformance, rejections, or complaints; no evidence of variability in final drug product attributes related to excipient	Minimal experience with manufacturer, and/or minor delivery issues	No experience with manufacturer, or manufacturer has a history of delivery issues; evidence of excipient variability; previous or outstanding complaints and impact on final product attributes
RF S_3: Organizational Volatility of the Excipient Manufacturer	Stable organization, low turnover in key positions	Limited knowledge of organizational volatility of supplier or manufacturer	High turnover in key positions
RF S_4: Excipient Manufacturer QMS Performance History (Customer complaint, changes, audits, ability to trust tailgate samples, etc. — oversight knowledge)	QMS works well and has proven effective; few or no regulatory observations; any findings are Minor; prompt audit response, CAPA on target per schedule	QMS is defined but there are some weaknesses in quality system	Significant quantity and/or severity of regulatory observations; adverse regulatory status (i.e. consent decree, severe or multiple warning letters, or official action indicated); area of special concern, critical or major findings, past due audit response, and/or CAPA items outstanding
RF S_5: Packaging Suitability (Ability to protect excipient from damage, e.g., fraud, moisture, heat)	Packaging is qualified and supported by scientific technical studies; tamper-evident packaging used	Data supporting suitability of the packaging is questionable	Packaging suitability is not supported by technical evidence; no tamper-evident seals
RF S_6: QMS Standards and Certification, Personnel and Training, Premises and Equipment, Documentation System and Records Management, Materials and Supplier Management, Quality System Elements	Established and implemented QMS that is well documented; certifications from regulatory bodies or other standardization organizations.	QMS implemented; no certification	Minimal evidence of QMS implementation; lacks documented supplier qualification program for suppliers of excipients, components, and finished products (as relevant); lacks service-level agreements to enforce quality control and related supplier programs; lacks transportation provider auditing or monitoring program (if relevant); lacks a documented process or does not enforce the process for management of production waste or damaged or expired product
RF S_x: Add according to specific need for the excipient in final drug product	To be defined	To be defined	To be defined

Each selection will result in a value of 1 (LOW), 3 (MEDIUM), or 5 (HIGH) for risk.

The risk score for the supply chain is calculated as the sum of the risk scores for each of the six identified risk factors (**RF S**)[3], divided by the number of risk factors. The resulting SC_{RPN} will be in the range of 1 (**Low**) to 5 (**High**).

$$Supply\ Chain\ Risk\ (SC_{RPN}) = \frac{RF\,S_1 + RF\,S_2 + RF\,S_3 + RF\,S_4 + RF\,S_5 + RF\,S_6}{6}$$

[3] Users would adjust the calculation as necessary to reflect additional risk factors that they have identified for consideration or to weight the individual risk factors based on the company's analysis and risk tolerance.

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The resulting risk score (**SC_{RPN}**) across the excipient risk factors **(RF S)** will be used, together with the overall score for the excipient described above, to evaluate the suitability and acceptability of the current supply chain setup.

6.5 Risk Evaluation

The excipient risk score and the supply chain risk score are then evaluated together using a risk matrix approach **(Figure 6.5-1),** based on the excipient risk score (**E_{RPN}**) on one side and the supply chain risk score (**SC_{RPN}**) on the other side, to create an overall risk score. **Figure 6.5-1** highlights one section of the complete risk model in **Figure 6.0-1.**

This final evaluation provides an overall view of the risk question "Is the excipient fit for use?" based on the strength of the manufacturer's and supply chain's GMP (or capability to control risk) and the risk related to the excipient's use in a drug product. The higher the risk related to the excipient's use and function, the greater the need for risk-controlling capability in the supply chain.

Because risk acceptability and the need for risk mitigation can vary significantly from company to company, the matrix and the associated cut offs showing risk acceptability should be tailored to the individual company's risk tolerance based on the drug products it manufactures.

Two options for the matrix are described and illustrated in **Section 6.5.1** and **Section 6.5.2**. To create the matrix using either option, the user first must understand all potential values for **E_{RPN}** and **SC_{RPN}**. When calculating **E_{RPN}** as described in **Section 6.3.2**, and **SC_{RPN}**, as described in **Section 6.4,** the result for each will be a number between 1 and 5 where a result of 1 is considered **Low**, and 5 considered **High**. In most cases, the **E_{RPN}** will lie somewhere between those two extremes, and will not necessarily be an integer. Further the scale is not linear and should not be treated as such.

6.5.1 Risk Matrix—Option 1: Calculating Combined Risk Score ($Overall_{RPN}$)

In Option 1, the combined risk score (**$Overall_{RPN}$**) is calculated by multiplying the **E_{RPN}** by the **SC_{RPN}**:

$$Overall_{RPN} = E_{RPN} * SC_{RPN}$$

The value of **$Overall_{RPN}$** indicates how well the risk is controlled and whether additional action is needed, as illustrated in **Table 6.5.1-1.**

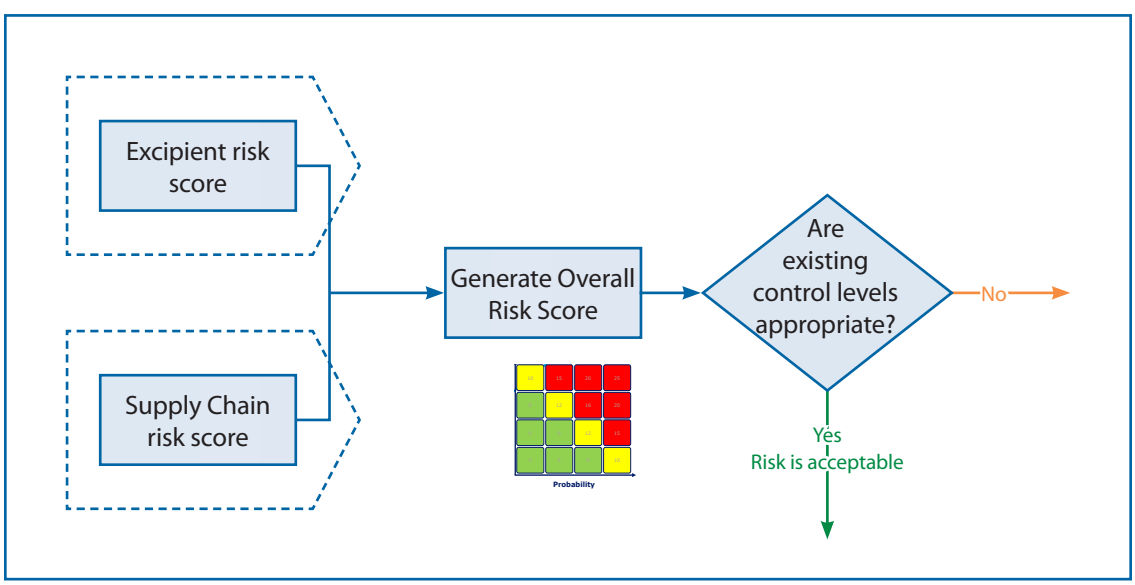

Figure 6.5-1 Developing the Matrix of Combined Excipient + Supply Chain Risk Evaluation

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Table 6.5.1-1 Risk Matrix to Evaluate Combined Risk Score for Excipient and Supply Chain

Supply Chain Risk (Low → High)	Excipient Risk 1 (Low)	2	3	4	5 (High)
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

From this combined risk score, the manufacturing authorization holder can determine whether additional actions are necessary. **Table 6.5.1-2** provides an example of a decision table specifying potential actions based on the **Overall**$_{RPN}$. **Section 6.6** discusses the selection of additional controls, if necessary.

Table 6.5.1-2 Decision Table to Determine Risk

Overall$_{RPN}$ > 10	Unacceptable – Residual risk is High	Additional actions to mitigate risk must be implemented
4 ≤ Overall$_{RPN}$ ≤ 10	As low as acceptable – Residual risk is Medium	Additional actions to mitigate risk should be considered, and acceptance justified
Overall$_{RPN}$ < 4	Acceptable – Residual risk is Low	No additional actions needed to mitigate risk to an acceptable level

6.5.2 Risk Matrix—Option 2: Calculating Final Risk Score

Option 2 for the risk matrix allows for additional customization by the manufacturing authorization holder. For instance, the manufacturing authorization holder may weight the factors according to its internal evaluation and controls. In addition, the manufacturing authorization holder may calculate and consider additional scores for other supply chain factors.

The final risk score is calculated by again using **E**$_{RPN}$ and **SC**$_{RPN}$. However, in calculating the final risk score, the manufacturing authorization holder can apply different weights to the two factors, depending on the manufacturing authorization holder's needs. For instance, if the manufacturing authorization holder has reason to believe that the excipient risk score should be given greater consideration, the excipient risk score could be given double the weight of the supply chain risk score in the final calculation.

The final risk score calculation is shown in **Figure 6.5.2-1**, where **n** is the sum of the weights in the numerator. If the excipient risk score is given a weight of 2 and the supply chain risk score is given the weight of 1, then n = 3.

Table 6.5.2-1 provides examples of the calculation of final risk score, each of which results in a scoring of LOW, MEDIUM, or HIGH. This final risk score guides the manufacturing authorization holder in determining whether additional actions are necessary.

$$\textit{Final Risk Score} = \frac{\text{[wt (Excipient Risk Score) + wt (Supply Chain Risk Score)]}}{\text{n}}$$

Figure 6.5.2-1 Calculating a Final Risk Score

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Table 6.5.2-1 Calculated Final Risk Score Examples

Excipient Risk Score (E_{RPN})	E_{RPN} weight	Supply Chain Risk Score (SC_{RPN})	SC_{RPN} weight	Final Risk Score
1.9	2	1.7	1	1.8
3.6	1	4.3	1	4.0
4.7	1	1.3	3	2.2
3.9	3	3.3	2	3.7
2.4	3	2.0	1	2.3

A manufacturing authorization holder also may make this calculation specific to its needs by considering, for instance, individual scores for each different role in the supply chain. That is, the manufacturing authorization holder may calculate a risk score for the original manufacturer of the excipient and another risk score for each player in the supply chain. In the calculation in **Figure 6.5.2-1**, the manufacturing authorization holder would use these individual mean risk scores in place of the more general supply chain risk score.

6.6 Implementation of Controls—Risk Mitigation

When the risk is found to be at an acceptable level, the risk assessment process is concluded. The identified controls should be documented, and the agreement with the manufacturer or supplier should ensure effective maintenance of those controls. This process is illustrated in **Figure 6.6-1.**

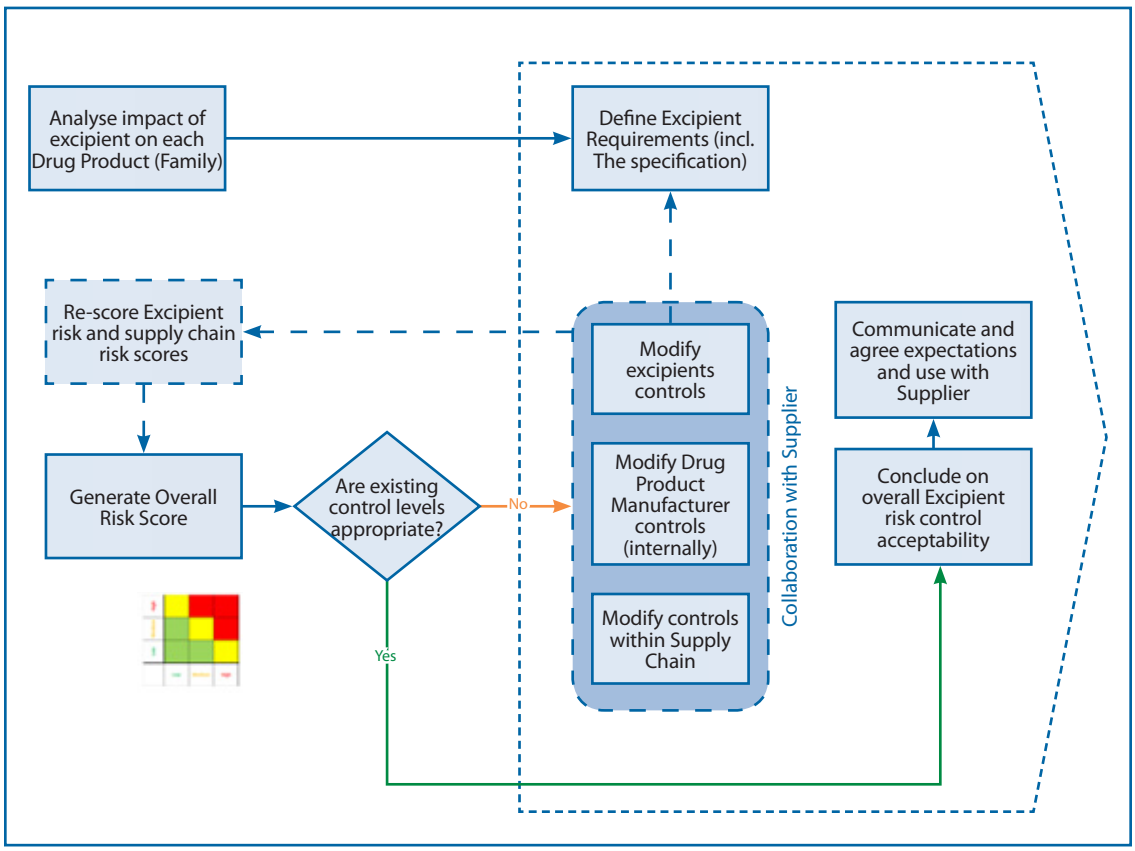

Figure 6.6-1 Identify Further Mitigation Actions to Reach Acceptable Risk Control, and Communicate

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If additional actions are needed, **Table 6.6-1** describes factors that should be considered. Depending on the specific excipient, other considerations might also need to be taken into account, as appropriate. Specific actions that might lower the residual risk can be found by deconstructing the $\mathbf{Overall}_{RPN}$ and investigating the risk scores for both the supply chain and the excipient that are driving the overall risk score.

Table 6.6-1 Factors to Consider in Determining if Additional Actions are Needed

Additional Drug Product Manufacturer Controls	Controls the manufacturer of the pharmaceutical product can introduce, for example: – Additional receiving inspection or testing – Special storage to avoid certain risks (e.g., degradation) – Additional processing of the excipient (e.g., filtration, irradiation)
Additional Excipient Controls	Additional Excipient Supplier Supply Chain Controls: – Change of GMP standard applied – Changes to specifications – Changes to packaging – Change of testing (e.g., every pack or every lot to ensure identity and homogeneity) – Microbiological testing – Regular verification of supplier's Certificates of Analysis – Introduction of new testing
Additional Excipient Supplier Supply Chain Controls	Controls the excipient supplier can introduce, for example: – Increased auditing by excipient user or third party – Changes to the excipient manufacturing process or controls – Higher demand for notification of changes – Increased oversight and control of shipments, tamper-proofing
Additional Excipient Manufacturing Controls	Additional controls the excipient supplier can introduce, for example: – Increased effectiveness of cleaning (microbiological risk reduction) – Sanitization (microbiological risk reduction) – Enhanced filtration of excipient, e.g., Technically Unavoidable Particle Profile (TUPPS) risk reduction – Altered processing conditions (by-product reduction) – Exclusion of rework (reduced unknown impurity risks) – Use of dedicated equipment – Introduction or change of in-process controls

The level of existing controls may exceed actual requirements. In this case, a company may consider reducing efforts in one area to focus on areas of greater risk, such as increasing audit frequency for excipients with high overall risk scores and decreasing audit frequency for excipients with low overall risk scores.

Confirmation that the additional controls have been agreed upon and implemented should be obtained from the excipient manufacturer or supplier and the drug product manufacturer, and should be documented, for example, in supply and quality agreements. Once risk control measures are in place, the risk assessment should be reviewed and rescored to determine the residual risk acceptability.

If the $\mathbf{Overall}_{RPN}$ is still unacceptable, further mitigation of the residual risks will be required. This cycle should be repeated until an acceptable residual risk has been obtained.

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6.6.1 General Considerations on the Appropriate GMP Required to Meet Risk Mitigation Level Determined by Risk Assessment

Figure 6.6.1-1 illustrates the extent and degree of risk mitigation that could be required based on the type of drug product and its use. This mitigation is defined by the excipient, how it is made, how it is supplied, why it is used, and how it functions in the drug product, as determined in the risk assessment. The degree of mitigation, that is, the reduction in overall risk that is necessary to reach a low residual risk level, increases as the different definitions of GMP are applied. The list of QMS standards and GMP definitions provided here is illustrative of the principle and is designed to help identify the GMP definition that reduces the risk to the desired level.

Once the required risk mitigation has been determined, the overall concept and necessary changes in requirements can be discerned, as discussed above in **Section 6.5.** As risks are mitigated, the overall risk assessment would be updated.

6.6.2 Applicability of GMP Standards

The EC Guidelines indicate the minimum GMP principles the manufacturer should implement in its quality and GMP management systems. These principles are embodied in most, if not all, GMP-related QMS and the standards listed in **Table 6.6.2-1.** Other standards providing the required degree of risk mitigation may also be available. (**Table 6.6.2-1** includes comments about the definitions of GMP in each.)

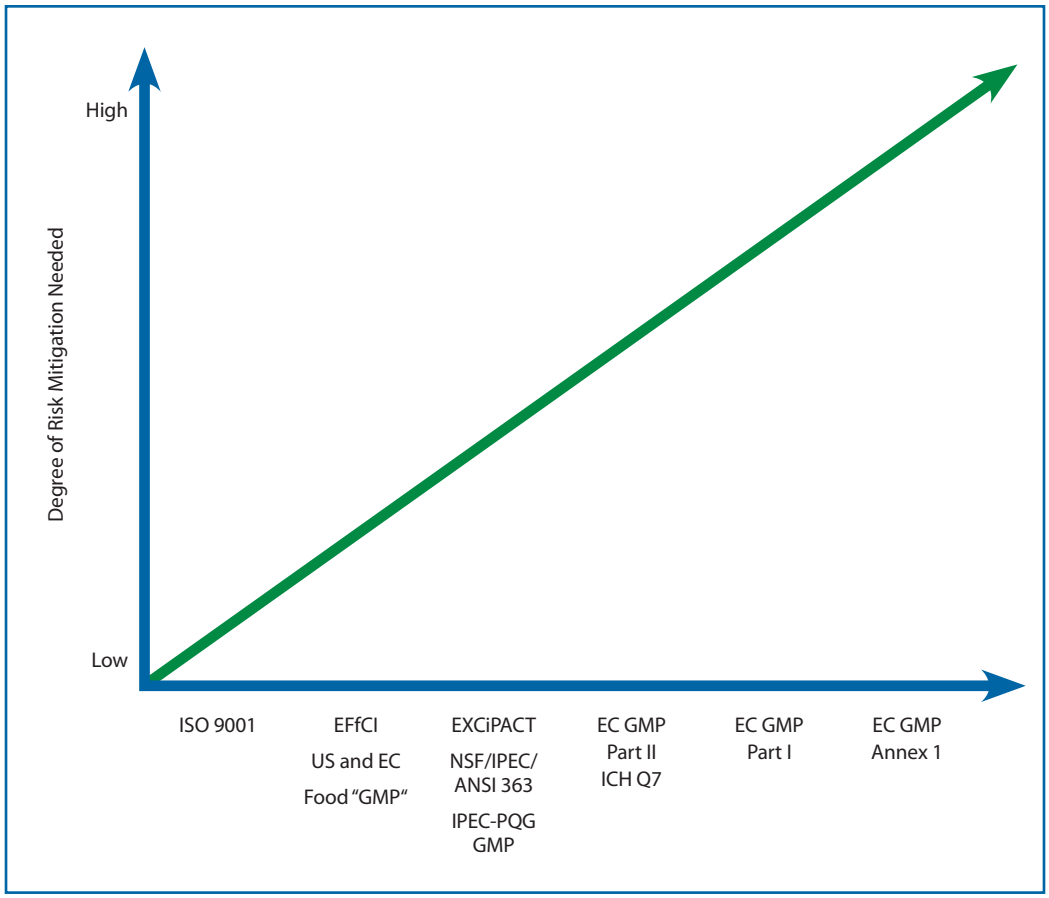

Figure 6.6.1-1 Degree of Risk Mitigation Required Based on the Nature of the Manufacturing Controls Required for the Excipient

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Table 6.6.2-1 GMP Standards by Standard Organization

Standard	Comments
EC Food "GMP" ***(20)*** U.S. Food GMP	Where both apply, a prerequisite QMS, in combination with hazard analysis and critical control point (HACCP) principles.
EC GMP Part II (Eudralex Vol. 4, Part II ***(21)***; ICH Q7 ***(22)***)	This is the GMP standard for active pharmaceutical ingredients.
EC GMP Part I (Eudralex, Vol. 4, Part I ***(23)***)	This is the GMP standard for drug products.
EC GMP Part I and Annex 1 (Eudralex, Vol. 4, Part I ***(23)*** and Annex 1 ***(24)***)	This is the GMP standard for sterile materials.
EFfCI GMP for Cosmetic Ingredients ***(25)***	This annex to ISO 9001 relies on the basic GMP principles in the EC Guidelines relating to risk assessment for excipients. The focus is on purity and contamination controls in manufacturing operations. Manufacturers must apply both ISO 9001 and EFfCI certificates simultaneously.
EXCiPACT™ GMP for Pharmaceutical Excipients ***(13)***	This annex to ISO 9001 has been designed for pharmaceutical excipients. It is based on the well-known and accepted *IPEC-PQG GMP Guide*, so it fully includes the principles of GMP in the EC Guidelines. Manufacturers must apply both ISO 9001 and EXCiPACT™ certificates simultaneously.
IFAC Quality Systems, Food Safety and Good Manufacturing Practices Guide for Food Additives and GRAS Substances ***(26)***	This guide provides guidelines on principles applicable under US requirements for food additives and generally recognized as safe (GRAS) ingredients, but also is intended to provide a unified approach for use by the international food additive industry.
IPEC Japan Self-imposed Standard of GMP for Pharmaceutical Excipient ***(18)***	This guide is based on the Joint IPEC-PQG GMP Guide (2006 version) with some modifications in view of the circumstances of Japan.
Joint IPEC-PQG GMP Guide ***(15)***	This is a guide and an audit standard. It was designed around the principles in ISO 9001 and ICH Q7 and, therefore, encompasses all GMP principles. A manufacturer compliant with this guide would have an equivalent level of risk mitigation as EXCiPACT™ and NSF/IPEC/ANSI 363.
ISO 9001 ***(27)***	This establishes a basic QMS that has the capability of delivering consistent quality materials. In many instances, the needed GMP details may be absent, so the risk mitigations may lack a high degree of assurance.
NSF/IPEC/ANSI 363 ***(14)***	This standard, for all practical purposes, is the same as EXCiPACT™. It is designed for manufacturers who do not apply ISO 9001.
USP General Chapter <1078> ***(17)***	This General Information Chapter discusses Good Manufacturing Practices for Bulk Pharmaceutical Excipients

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7.0 Ongoing Review—Formalized Risk Assessment Lifecycle

Once the excipient risk assessment is formally in place, it will be used as living data and will contribute to a knowledge base through ongoing risk review. **Figure 7.0-1** shows the overall generic risk model with respect to risk review only, for simplification. Risk review generally will be conducted periodically (i.e., monitoring) or as a result of triggering factors.

7.1 Triggers for Risk Assessment Review

Trending of excipient and drug product quality plays a key role in the risk assessment to ensure science- and data-based assessments. Key performance indicators or other metrics should be monitored on an ongoing basis to demonstrate the effectiveness of the risk controls. That monitoring should highlight any triggers that indicate the need to reassess the risk and, possibly, revise the assessment.

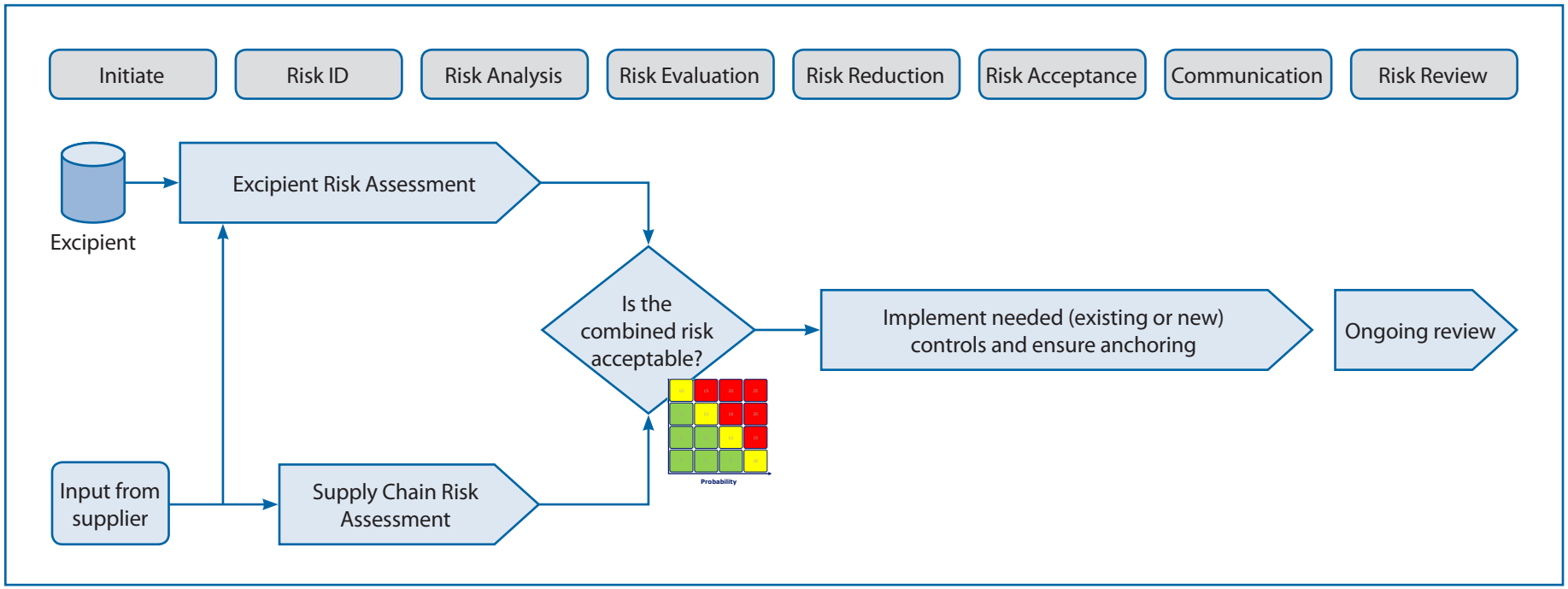

Figure 7.0-1 Simplified Overview of the Generic Lifecycle Model

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Internal procedures should define the periodic review and revision process and list the events that would trigger a reassessment, such as:

- New product
- New formulation of existing product
- New manufacturer/supplier
- New excipient
- Change in notification from manufacturer/supplier
- Change in supply chain
- Change in GMP certification status or GMP status of manufacturer/supplier
- Change to site
- Geopolitical/socioeconomic/business changes
- Regulatory changes
- Pharmacopeial changes
- Quality events, industry-wide (e.g., TSE)
- Shifts/changes in trends
- New safety information

When triggering events occur, both the event and the existing excipient risk assessment should be reviewed, and a decision should be made about the need to revise the excipient risk assessment based on the new knowledge. The process, decision, and rationale should all be documented. Updates to the risk assessment may lead to revised control decisions.

8.0 Benefits and Value / QRM / Knowledge Management

A global excipient risk assessment throughout the drug product lifecycle delivers the following advantages:

- Documented knowledge, in a structured manner, about the excipient supply chain
- Ability to rapidly and effectively evaluate the requirements for new products that use existing or new excipients
- Identification of supply chains that present lower risks and preferred supply chains with which to build relationships
- Focus on specific risk mitigation oversight activities for both manufacturer or supplier and user
- More detailed product knowledge and increased understanding of how the risks from excipients may affect product quality
- Formal assessment and database of risk review triggers, such as major changes and critical investigations
- Knowledge-based and proactive decision-making
- Provides a common language and approach for clear communication with stakeholders throughout the supply chain
- Opportunities for knowledge-sharing and comparative assessments between manufacturers or suppliers and end users

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10.0 Annex I: Real World Examples of Risk Assessments

The generic model proposed in **Section 6.0** was developed based on real-world risk assessment processes donated by several marketing authorization holders. Many risk assessment approaches used some form of risk ranking and filtering approach—conducting assessments in two separate streams that can then be combined in an overall risk score.

One stream would provide a risk profile for the excipient itself, based on its use in the finished product and any intrinsic risk factors. The second stream would generate a risk profile for the supply chain, including the excipient manufacturer, suppliers, brokers, and any other players, as well as the strength of their quality management systems (QMS). The two streams would then be combined into one risk evaluation to determine whether the overall risk is at an acceptable level or additional controls are needed.

10.1 Example 1: Company U

Company U is a global manufacturer of various pharmaceutical dosage forms. The company established a Microsoft® Excel® tool to be used as a corporate standard.

When multiple manufacturing sites use the same excipient provided by the same excipient manufacturing site, the risk assessments can be shared across all sites (with appropriate site-specific modifications to accommodate excipient usage, supply chain complexity, and other factors).

First, basic information about the excipient and the manufacturer is collected. Then, the collected information is consolidated to produce an overview.

For each section of the model, the parameters to evaluate are defined (see **Figure 10.1-1),** each parameter is described in detail, and possible scenarios are developed. The severity of each parameter is assigned a standard value. In conducting the evaluation, Company U determines the actual probability of each parameter.

The analysis is conducted as follows:

- **Excipient and excipient usage risk score —** Each of the identified excipient and excipient use risk factors is evaluated based on the actual origin or usage of the excipient and the actual supply chain to determine the probability of risk **(P).** The predefined weight, or severity **(S),** is used to calculate the risk score for each parameter **(P × S).** Then, the overall excipient risk score is calculated as the average across the parameters.
- **Combined material risk score —** A combined material risk score is calculated by adding the excipient risk score and the excipient usage risk score.
- **Manufacturer risk score —** The list of risk factors to be considered is evaluated based on the actual performance and QMS of the manufacturer. The predefined weight (**S**) is used to calculate the risk score of each parameter **(P × S).** Then, the overall manufacturer risk score is calculated as the average across the parameters.
- **Gap analysis and actions —** The combined material risk score and the manufacturer risk score are evaluated together using a risk matrix to identify the "gap level," as illustrated in **Table 10.1-1.**

  Where a gap exists between the manufacturer risk score and the combined material risk score, the gap must be evaluated to determine the appropriate remediation.

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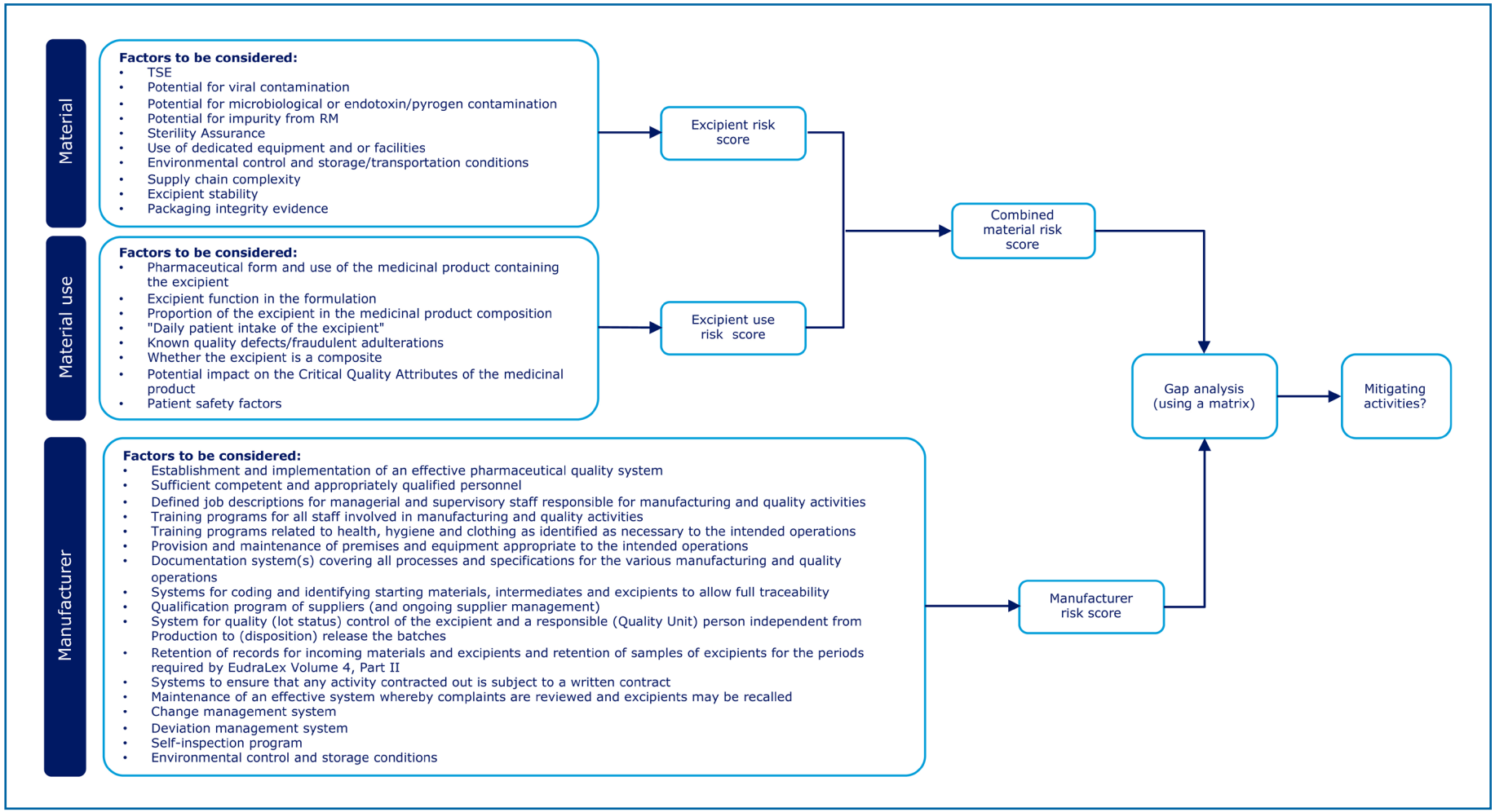

Figure 10.1-1 Risk Assessment Model for Company U

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Table 10.1-1 Gap Analysis Matrix Used to Determine Severity of Risk

	Excipient & Excipient Usage Risk (Combined Risk)			
	Risk Level	Low	Medium	High
Manufacturer Risk	High			
	Medium			
	Low			

10.2 Example 2: Company V

Company V is a global manufacturer of pharmaceutical and biopharmaceutical solid oral dosage forms and lyophilized injectables. To a large extent, its products are manufactured through contract manufacturing organizations (CMOs).

The risk model **(Figure 10.2-1)** was created to meet the European Commission *Guidelines on The Formalised Risk Assessment for Ascertaining the Appropriate GMP For Excipients of Medicinal Products For Human Use* (EC Guidelines), and the risk factors to be considered reflect those in the guidelines ***(1).*** The model was implemented in a global SOP that applies for all sites manufacturing commercial products and, going forward, for new products, including biologics and other type of products.

This model was extended for products during their transfer into commercial production. The quality risk assessment tool is applied based on risk ranking and filtering, evaluating the risk factors in two groups:

- Multiple criteria analysis
  - Function and use of excipient (→ Score Excipient)
  - Manufacturer (→ Score Manufacturer)
  - Suppliers (→ Score Supplier)
- Risk score for each excipient (LOW, MEDIUM, or HIGH)
  - Independent risk factors
  - Risk ranking and weighing
  - Overall weighing

The three single scores—Function and Use, Manufacturer, and Suppliers—were combined by multiplication into a weighted RPN score. Acceptability was evaluated based on the RPN value, and mitigation actions were considered according to the overall risk score (LOW, MEDIUM, or HIGH).

- No major issues were found for existing products, though some minor changes in audit strategy and other controls were instituted as a result.
- One benefit of performing the risk assessment was obtaining the collected knowledge and an overview of the supply chains.

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- Company V has found that responsibilities for information-gathering need to be changed in the global SOP. Currently, the global support function gathers the needed information from different stakeholders. Using one point of contact for each product is under discussion as an alternative.
- While the model has been well accepted, it has not yet been used as a living risk assessment as much as would be liked.
- The risk assessment is stored in the GMP documentation system. Software automatically triggers formal review and approval cycles globally. More flexibility to revise and update the risk assessment would be appreciated, especially in cases where the risk assessment is completed before the drug product is commercially available. The company will introduce a new electronic tool to serve this purpose.

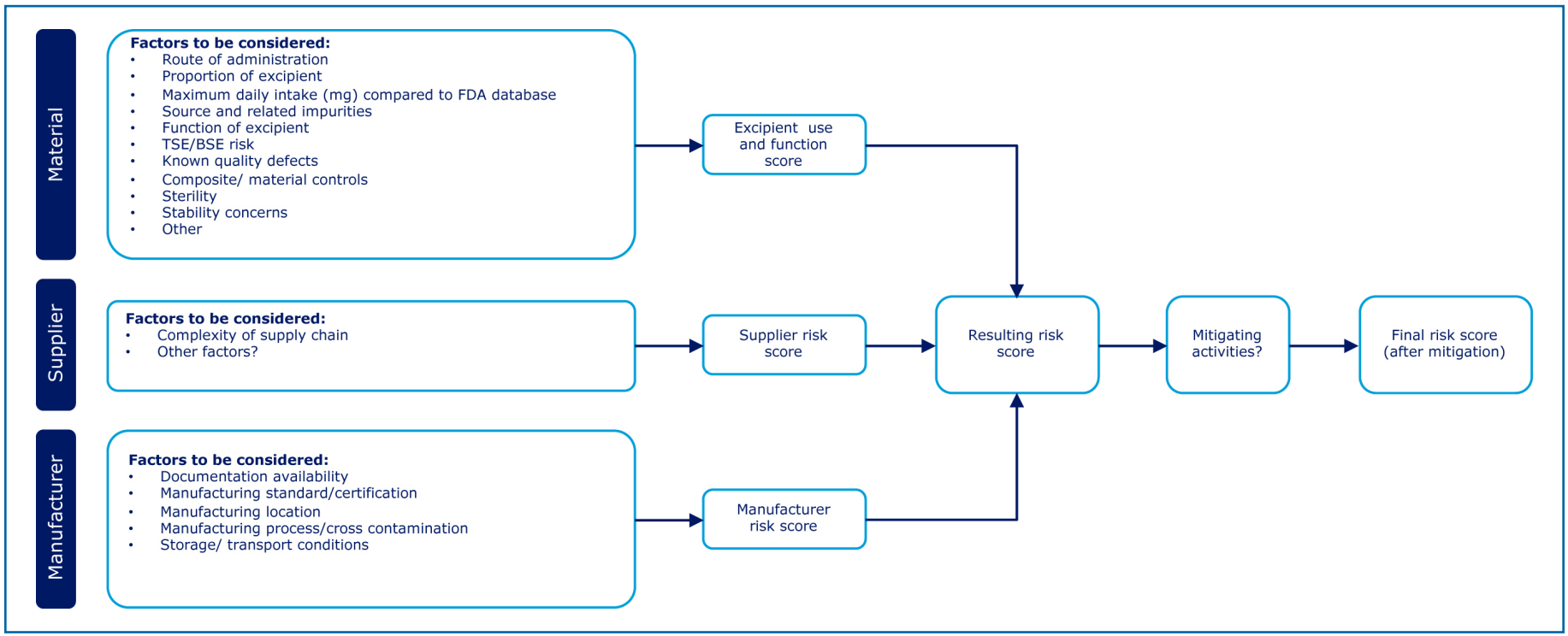

Figure 10.2-1 Risk Assessment Model for Company V

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10.3 Example 3: Company W

Company W is a global manufacturer of pharmaceuticals and devices, primarily parenteral products.

As part of the material assessment, the company examines the correlation of impact between parameters (attributes) describing the excipient and the CQAs of the product in which it is an ingredient. An excipient assessment provides input to material specifications, as illustrated in **Figure 10.3-1.** In this way, the company evaluates the impact of the excipient on the finished drug products. Attributes are derived by ideation from a defined list of potential risk sources. Attributes with impact are included on the specification, and limits or other controls are defined after analyzing the possible impacts of variation in the attribute. This method uses a qualitative assessment with a simple +/- score for impact combined with text rationales and references describing the underlying science. The impact is evaluated considering the amount of material to be used and its function in the formulation.

The result of the assessment is used to define the excipient's specification based on the specific use in a specific drug product. An assessment is made for each use in a drug product.

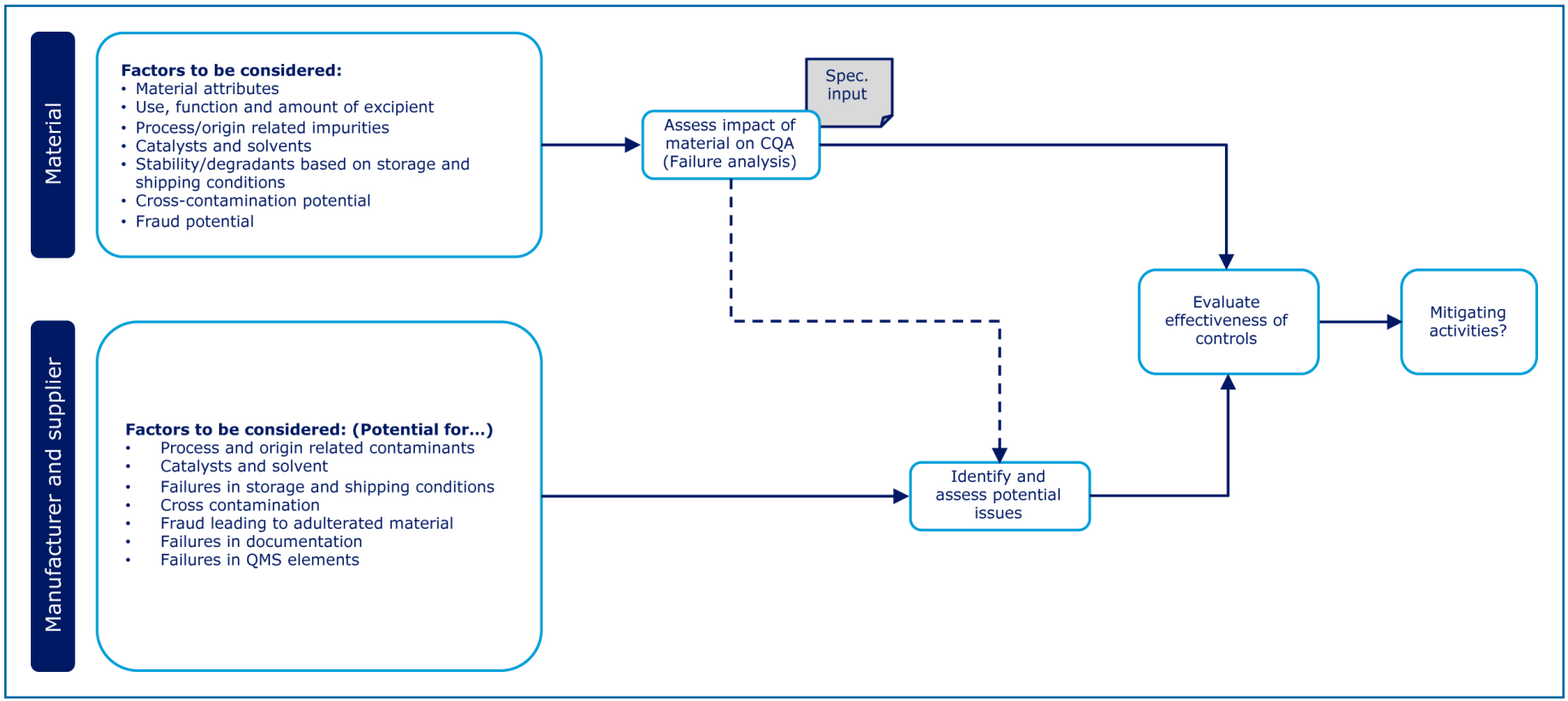

Figure 10.3-1 Risk Assessment Model for Company W

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An assessment of the supply chain is carried out in parallel with the material assessment. The supply chain—including manufacturer, retailers, brokers, and others—is evaluated using a predefined list of risk elements, brainstorming possible relevant issues in the specific supply chain for the particular excipient. The identified risks are then evaluated qualitatively for impact, likelihood, and controls in place.

If a risk becomes evident, the company identifies any additional controls that would reduce the risk to an acceptable level.

10.4 Example 4: Company X

Company X expects its suppliers to comply with the IPEC-PQG *Joint Good Manufacturing Practices Guide* ***(2),*** EXCiPACT™ Standard ***(3)***, or NSF/IPEC/ANSI 363 ***(4).*** The excipient risk management process considered input from the manufacturer qualification process, audits, relationships, testing, and outside intelligence. The company also worked with its CMOs to align expectations between them and the manufacturing authorization holder and rationalized scores. The actual process is owned by the company's organizational product quality management unit and follows the model illustrated in **Figure 10.4-1.**

The product quality manager (PQM) is responsible for pulling together the information and subject matter experts from across Company X to provide the data needed to complete the spreadsheet. Much of the information comes from the company's audit team. The model allows for rationalization across materials and suppliers to ensure that scoring is consistent and can be readily justified.

Company X maintains an SOP that describes the overall process, including triggers for revisiting a risk assessment. Because the PQMs also are involved in the change control process for company materials, they have firsthand knowledge of any changes that would require a review of the excipient risk assessment. As part of this process, Company X has already made several revisions to some excipient risk assessments.

Initially, there was some resistance to the concept and process. Now that the system is in place, however, staff have noticed that it is easier to find information without going into several systems or speaking to several departments.

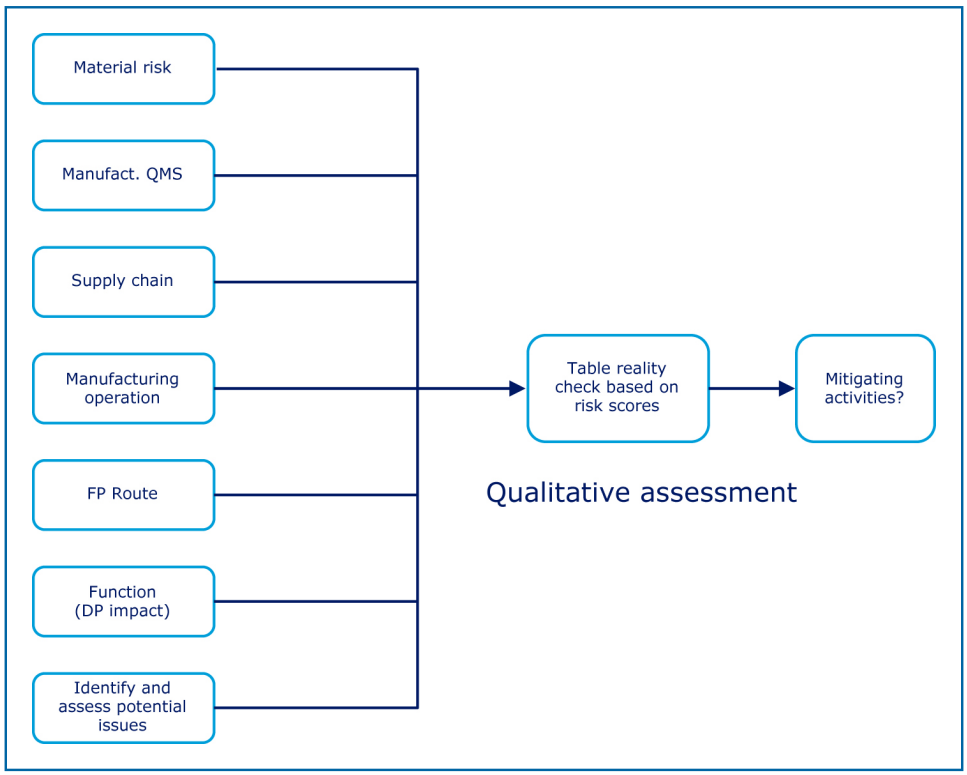

Figure 10.4-1 Risk Assessment Model for Company X

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10.5 Example 5: Company Y

The model illustrated in **Figure 10.5-1** applies to finished products for human use that Company Y manufactures for European markets.

Company Y manufacturing sites completed risk ranking of each excipient from each manufacturer used at their sites.

A risk ranking tool is used to evaluate each excipient from each manufacturer. A weight of 1 (LOW), 3 (MEDIUM), or 10 (HIGH) is assigned for each of six criteria and, from that, an overall risk ranking of the excipient (LOW, MEDIUM, or HIGH) is calculated. Where the GMPs used to manufacture excipients are found to be inappropriate according to risk ranking, mitigation measures are implemented. A risk analysis tool for manufacturer audit frequency is then used to assess the risk and determine audit frequency. Based on the outcome, a second risk assessment could be performed to determine if (1) additional mitigation measures should be implemented, or (2) use of the excipient from that specific manufacturer should be discontinued. If Company Y decides to continue to use the excipient after implementing additional mitigation measures to reduce the remaining risk to an acceptable level, the company documents this decision in a "statement to maintain business."

Company Y periodically reevaluates risk profiles during the review of a manufacturer's performance. A risk assessment is reevaluated in cases of a significant quality event, e.g., significant change by manufacturer, serious GMP deficiencies, or loss of GMP certification.

- The model is simple to implement.
- The result of a risk assessment sometimes varies between sites. However, after implementing a control strategy, the sites could reach the same decision to continue use of the excipient from that manufacturer.
- Use of this model aids Company Y in implementation and modification of control measures to reduce risk to an acceptable level.

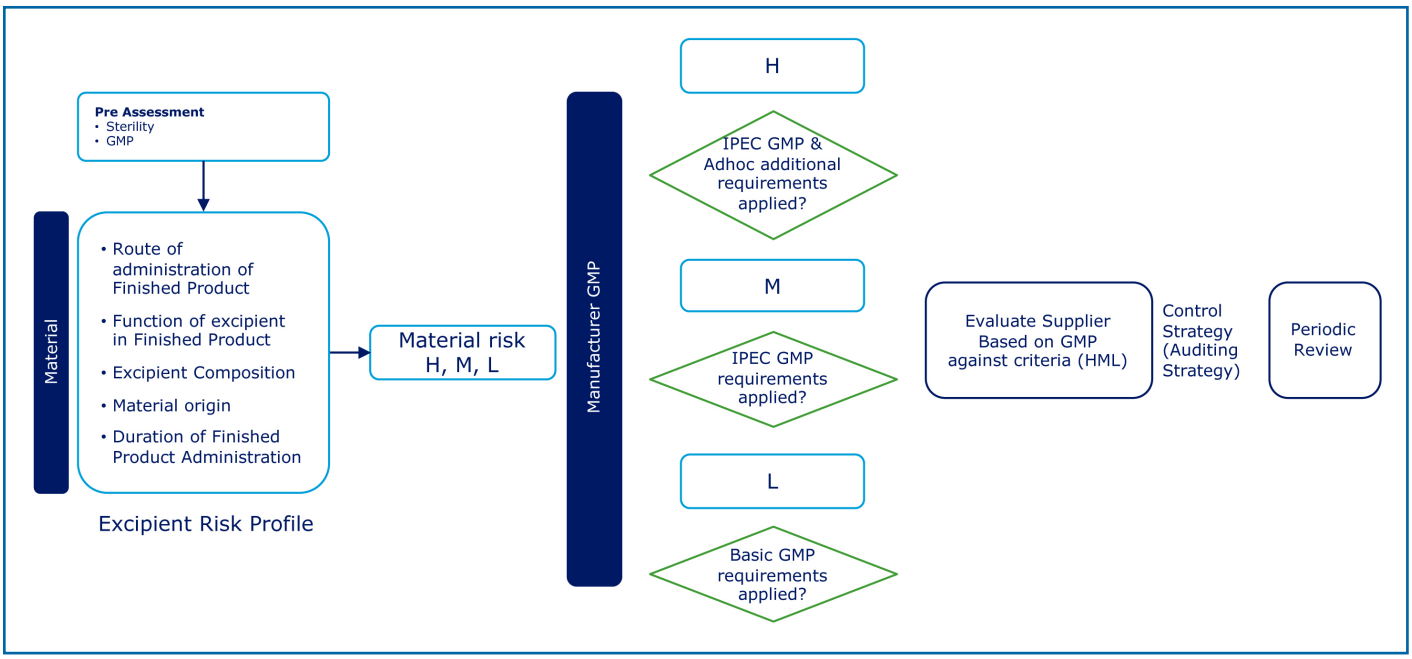

Figure 10.5-1 Risk Assessment Model for Company Y

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10.6 Example 6: Company Z

Company Z's model (**Figure 10.6-1**) was designed as a centralized process focused on highlighting the inherent risks and control strategies associated with an excipient, regardless of the manufacturer. The process is also designed to provide the user with the minimum quality standard expected for the manufacture of the excipient and its associated key risks. These risk and control strategies provide input for the gap analysis and, subsequently, to the quality risk assessment performed on supplier sites.

Company Z established a corporate procedure that outlines the requirements for the excipient risk assessment process and its linkages to the overarching supplier quality management process.

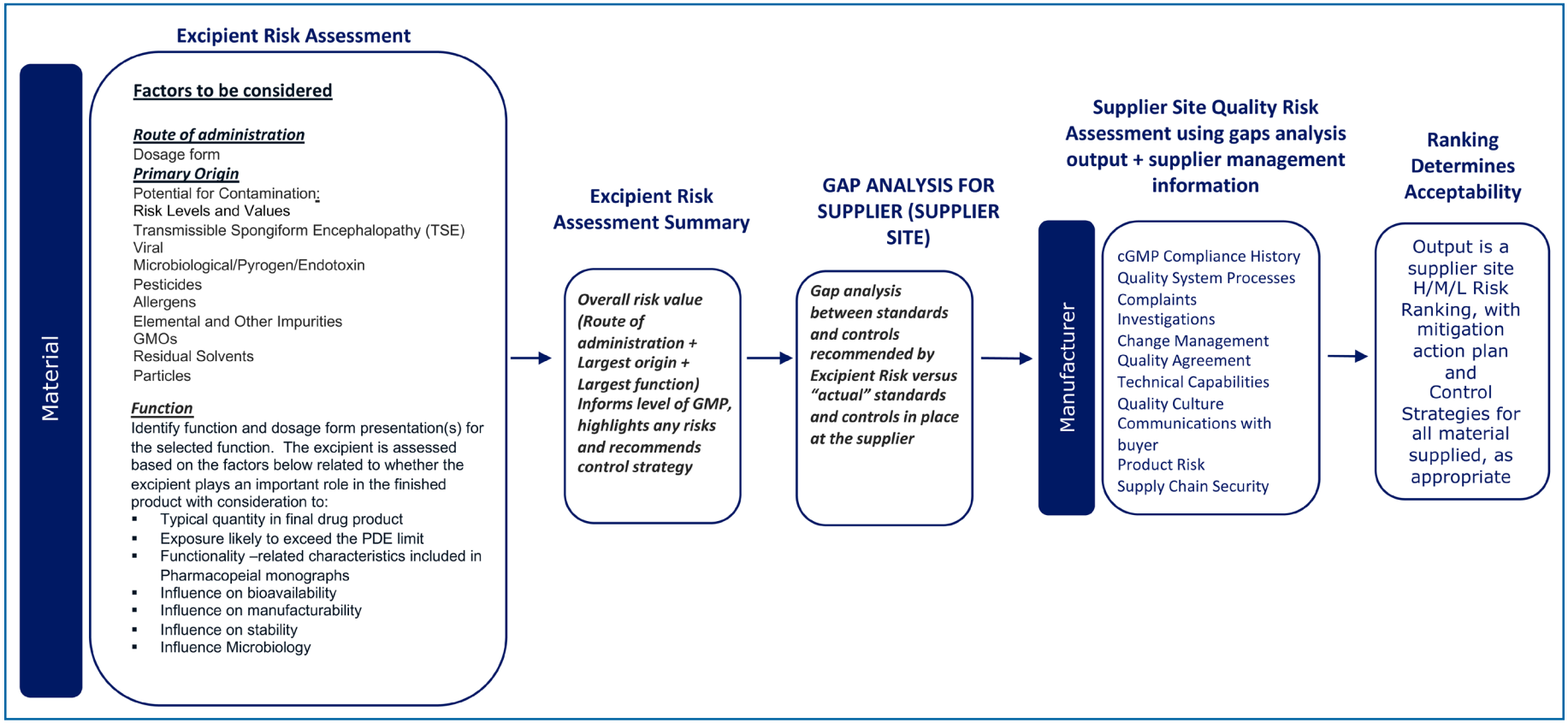

Figure 10.6-1 Risk Assessment Model for Company Z

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The first part of the "end-to-end process excipient risk assessment" determines the control strategy and standards that are established by the centralized excipient risk assessment team. In the second part, the quality management group of the global supplier uses the output of the risk assessment to conduct individual supplier site quality risk assessments. The information from the individual supplier site quality risk assessments is shared with the formulation sites through a vendor assurance database system, highlighting any changes in supplier risk ranking via change control. The excipient risk assessments are reviewed periodically and any changes in inherent risk are brought to the attention of the global supplier's quality management team. In addition, any quality event that impacts the excipient quality, whether internal or external, results in a review of the initial excipient risk assessment.

After the need for an excipient risk assessment is identified, the centralized, cross-functional excipient risk assessment team performs the assessment, which focuses entirely on inherent risk. Supplier risks are not considered at this time. An electronic tool is used to perform the assessment, and the output is used to determine the appropriate quality level for the excipient and any relevant excipient control strategy.

Each supplier quality manager from Company Z's global supplier quality team then performs a gap analysis of the excipient supplier quality standard against an assessment of defined standards and excipient control strategies for supplied excipients. This identifies any gaps for the supplier. All gaps are challenged to ensure their validity and valid gaps are used as input, along with other supplier quality management intelligence, in performing the supplier site quality risk assessment. This final assessment generates a supplier risk ranking (**LOW**, **MEDIUM**, or **HIGH**) which, in turn, determines the quality oversight to be applied.

The maintenance of the excipient risk assessment is managed under the periodic review process outlined in the corporate standard on excipient risk management. Quality events that impact excipient quality trigger an ad hoc excipient risk assessment review.

Company Z implemented this process in March 2016 in line with the EC Guidelines.

- A key benefit of conducting the assessment is consistency in managing excipients and suppliers. The centralized approach provided a concise framework for consistent input into the initial steps of the process. The implementation formalized the process and improved the documentation of information, supporting existing supplier risk assessment processes.
- Grouping excipient grades into categories aligned with the route of administration, origin, and function has reduced the number of assessments required.
- Conducting a formalized assessment process provides clarity to sourcing partners, both internal and external.
- The risk assessment acts as an effective standard for comparing and assessing supply chain risks.

11.0 References (for Annex 1)

1. European Commission. *Guidelines of 19 March 2015 on the Formalised Risk Assessment for Ascertaining the Appropriate Good Manufacturing Practice for Excipients of Medicinal Products for Human Use*. European Commission: Brussels, 2015.
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3. EXCiPACT asbl. *EXCiPACT: Annex to ISO/IEC 17021-1:2015 Additional Requirements for Conformity Assessment Requirements for Certification Bodies and Auditors Revision 2017*; EXCiPACT: Ann Arbor, Mich., 2017.
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